

Region-Aware Optimization of Hierarchical Bloom Filters for Privacy-Preserving Face Membership Verification

By

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Abstract

Face membership verification asks whether a query face belongs to an enrolled group. Unlike face identification, the system does not need to return a named identity. This is useful for access control, duplicate-enrollment checks, watch-list membership, and similar applications. The stored representation should be compact, quick to query, and still separate genuine and impostor samples. Since biometric information cannot be replaced like a password, reducing direct exposure of facial data also matters.

This thesis proposes Landmark-Indexed Bloom Filters (LBFs), a region-aware extension of the Hierarchical Bloom Filter approach. Instead of treating the face as one global binary pixel sequence, LBF stores compact binary codes at selected facial landmarks. Each code is tied to a specific landmark and queried through its own Bloom filter. The main contribution is this change in organization, where the Bloom-filter structure follows facial regions rather than positions in a flattened image.

Three pipelines are evaluated. Pipeline 1, the Baseline HBF, follows the original hierarchical pixel-block design and reaches an equal-dataset macro AUC of 0.6754. Diagnostic experiments show that the upper hierarchy levels add little ranking information, many pixel positions are shared across identities, and useful facial evidence is not distributed evenly across the face.

Two LBF variants are then developed. DCT-LBF uses low-frequency DCT codes at selected landmarks and reaches a macro AUC of 0.8675. Deep Spatial LBF keeps the same 48 landmarks and 72-byte template, but replaces the DCT representation with identity-trained ArcFace spatial features, reaching a macro AUC of 0.9334. Compared with the baseline, both LBF pipelines reduce template size by 88.3% and total HBF-bank storage by about 71.3%.

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List of Abbreviations

Abbreviation	Meaning
AC	Alternating-current DCT coefficient (a nonzero-frequency coefficient)
AUC	Area Under the Receiver Operating Characteristic Curve
BF	Bloom Filter
BSIF	Binarized Statistical Image Features
DC	Direct-current DCT coefficient (the zero-frequency coefficient)
DCT	Discrete Cosine Transform
FN	False Negative
FNV	Fowler-Noll-Vo Hash Function
FP	False Positive
HBF	Hierarchical Bloom Filter
HMAC	Hash-Based Message Authentication Code
HOG	Histogram of Oriented Gradients
IoT	Internet of Things
DCT-LBF	Discrete Cosine Transform Landmark-Indexed Bloom Filter
LBP	Local Binary Pattern
ONNX	Open Neural Network Exchange
PCA	Principal Component Analysis
LBF	Landmark-Indexed Bloom Filter
ROC	Receiver Operating Characteristic
SHA-256	Secure Hash Algorithm with a 256-bit output
TN	True Negative
TP	True Positive

Abbreviations and their meanings.

Chapter 1: Introduction

Face membership verification asks a narrower question than face identification: does a query face belong to an enrolled group? This type of decision can support access control, duplicate-enrollment checks, watch-list membership, and similar applications where the system does not need to return a named identity. The representation still must separate genuine and impostor queries while remaining compact enough for repeated storage and fast membership checks. Since biometric information cannot be reissued like a password [3], how facial information is represented and stored is also important. In this work, privacy-preserving membership verification is treated as a practical design goal. The evaluated banks store keyed, hashed membership structures instead of raw face images or directly searchable real-valued feature vectors. This does not represent a complete proof of irreversibility, unlinkability, or resistance to every possible attack.

Bloom filters are useful for membership testing because they require relatively little storage and support fast queries [1]. Their main limitation for biometric data is that the queried item may not be the same as the enrolled item. The original Hierarchical Bloom Filter (HBF) addresses this by dividing a binary face template into blocks of different sizes and combining the resulting block matches into a membership score [2]. Pipeline 1 in this study follows this design. An aligned face is converted into a 4,900-bit mean-thresholded pixel template, which is then divided into low-, middle-, and complete-template blocks.

The baseline was first audited before any changes were made. This was necessary because weak performance could come from the Bloom-filter configuration, an implementation problem, the hierarchy, or the binary representation itself. The diagnostic experiments showed that long blocks rarely survive changes between photographs of the same person. The upper hierarchy

levels also contributed little ranking information, while many positions in the global pixel representation were shared across different identities. These results suggested that increasing Bloom-filter size alone would not address the main limitation.

Based on this diagnosis, this thesis proposes Landmark-Indexed Bloom Filters (LBFs). Instead of treating the face as one global binary sequence, an LBF assigns a compact local descriptor and binary code to selected facial landmarks. Each code is stored and queried through the Bloom filter assigned to that landmark. A code around the left brow is therefore treated differently from the same code around the jaw. The main change is that the membership structure follows repeatable facial locations rather than arbitrary positions in a flattened pixel sequence.

Two LBF pipelines are evaluated using the same identity-disjoint fit, development, and evaluation protocol. Pipeline 2: DCT-LBF uses landmark-centered image regions and fixed low-frequency DCT codes. Pipeline 3: Deep Spatial LBF keeps the same landmark set, 72-byte template, regional backend, thresholds, and data partitions, but replaces the local DCT representation with a frozen identity-trained ArcFace spatial descriptor compressed using PCA. The main contribution of this work is the LBF framework and the evaluation of DCT-LBF and Deep Spatial LBF against the Baseline HBF. Chapters 2 and 3 describe the related work and experimental protocol, Chapter 4 presents the baseline diagnosis, Chapter 5 introduces the LBF framework and its two variants, Chapter 6 compares the pipelines, and Chapter 7 concludes the study.

Chapter 2: Literature Review

This chapter reviews the background most relevant to the membership-verification problem studied in this thesis. The discussion begins with biometric templates and membership verification. It then covers facial landmarks, binary representations, Bloom filters, and the Hierarchical Bloom Filter (HBF) framework. The later sections introduce DCT, ArcFace, and PCA, which are used in the two region-aware pipelines developed in this work. The chapter ends by identifying the representation and organization gap addressed by the proposed Landmark-Indexed Bloom Filter framework.

2.1 Biometric Membership Verification

A biometric system first needs a representation of the input sample that can be stored or compared later. A feature extractor converts a biometric sample into a template. A useful face template should preserve information that separates identities while reducing changes caused by image acquisition. It may contain pixels, handcrafted features, or features learned by a neural network. [3]

Within-person variation is the difference between photographs of the same person. Between-person variation is the difference between different people. A strong representation keeps the first small and the second large.

Face verification compares two templates. Membership verification only asks whether the query belongs to any enrolled identity, which allows the database to store a group-membership structure. [2]

The design must balance recognition, storage, and template protection. This is difficult because biometric data vary naturally and cannot be revoked as easily as a password [3].

2.2 Binary Templates

A binary template stores each feature as zero or one. Binarization reduces size and makes exact block comparison simple. A feature above a fitted reference can become one, while a feature at or below it becomes zero.

The published HBF compares each aligned pixel with a mean face and produces one bit per pixel [2]. This simple method can encode common structure: the same forehead positions may be bright, and the same eye-socket positions dark for many people. Stable bits are not necessarily identity specific.

Binarization also creates a hard boundary. A small change around the threshold can flip a bit. When several bits are hashed together, one flipped bit changes the hashed block [2]. The HBF manages this effect by using many blocks, but its performance still depends on the information and stability of the binary template.

2.3 Local Facial Descriptors and Landmark-Based Evidence

Before facial information can be compared across photographs, differences in translation, rotation, and scale need to be reduced. This thesis uses the dlib library for face alignment, localization and 68-point landmark prediction. The dlib frontal-face detector combines histogram-of-oriented-gradient features, a linear classifier, an image pyramid, and a sliding-window search to return a face bounding box [21]. The `shape_predictor_68_face_landmarks.dat` model then predicts points around the jaw, brows, nose, eyes, and mouth.

The predictor implements the cascaded ensemble-of-regression-trees method of Kazemi and Sullivan [4]. Facial landmarks also provide a natural way to test whether identity information is distributed evenly, which makes them an interesting choice for region-aware bloom filters.

2.4 Bloom Filters

Bloom filters were introduced as compact probabilistic data structures for membership testing [1]. A Bloom filter contains an array of m bits, initially set to zero, and uses k hash functions.

The same hashing procedure is used for both enrollment and query [2].

When an item is enrolled, the k hash functions map it to k positions in the bit array, and those positions are set to one. A query repeats the same process. If any required position is zero, the queried item was not inserted. If all required positions are one, the filter reports that the item may be a member.

The result is only a possible membership because different enrolled items can set some of the same positions. An item that was never inserted may still find all its required positions set to one.

This is the **Bloom-filter false positive**. Under the standard independent-hash approximation, the false-positive probability is

$$f \approx (1 - e^{-kn/m})^k, \quad (2.1)$$

where n is the number of enrolled items, m is the number of bits in the filter, and k is the number of hash functions. For a fixed m and n , the approximate optimal number of hash functions is

$$k = \frac{m \ln 2}{n}. \quad (2.2)$$

With this choice of k , the false-positive relation can be written as

$$FP_{BF} \approx 0.6185^{m/n}. \quad (2.3)$$

Equations (2.2) and (2.3) are used in the HBF design described by Shomaji et al. [2]. They show the main storage trade-off. For the same number of enrolled items, lowering the target false-positive probability requires a larger Bloom filter.

Two sizes are used throughout this thesis and should be kept separate. P is the number of bits in one biometric template, while m is the number of bits in one Bloom filter. These values describe different parts of the system. A change in the frontend representation can change P . The enrolled population and the selected false-positive target affect m .

Symbol	Meaning
d	Highest hierarchy level
e	Number of changed bits between two templates of the same identity
k	Number of hash functions used by one Bloom filter
l_i	Block length at hierarchy level i
m	Number of bits in one Bloom filter
n	Number of enrolled templates
N	Total number of Bloom filters in the HBF
p_t	Per-bit disagreement rate, e/P
P	Number of bits in one complete biometric template
w	Block-growth factor between hierarchy levels
τ	Decision threshold applied to the HBF match score

Table 2.1. Main HBF notation.

2.5 Hierarchical Bloom Filters (HBF)

A standard Bloom filter is most suitable when the queried item is exactly the same as the enrolled item. Biometric data do not normally satisfy this condition. Even two photographs of the same person produce slightly different templates. If the entire biometric template is treated as one Bloom-filter item, a change in only one bit can cause the query to miss.

Shomaji et al. divide the binary facial representation into blocks and store the resulting values across multiple Bloom-filter levels [2]. Their method uses separate enrollment and query procedures. For enrollment, the face image is processed and quantized before its blocks are added to the filter bank. For a query, the same preparation stages produce blocks that are checked against their assigned filters. Membership is returned when the number of successful checks reaches the selected threshold. The output indicates whether the face belongs to the enrolled group rather than identifying a particular person.

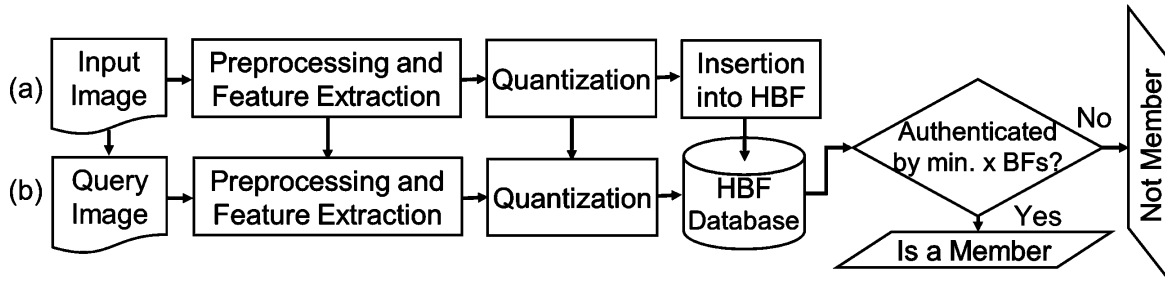


Figure 2.1. HBF enrollment and query workflow. Source: Shomaji et al. [2, Fig. 1]. © 2019 IEEE.

Enrollment places each block in the Bloom filter associated with its level and position. Query processing repeats the partition and checks every block in the corresponding filter. The membership score is the number of successful block-level tests [2].

The reason for using several block sizes is the trade-off between tolerance and selectivity. Short blocks have a better chance of remaining unchanged when a biometric sample varies. Longer

blocks preserve more neighboring information and are less likely to match an unrelated template by chance. Instead of requiring one exact whole-template match, the hierarchy obtains partial agreement from many exact block tests [2].

This behavior can be related to the bit disagreement between two samples. Let e be the number of positions that differ between two templates of the same identity and let P be the template length. Their bit-disagreement rate is

$$p_t = \frac{e}{P}. \quad (2.4)$$

If bit changes are treated as independent, a block of length l_i survives only when none of its bits changes. Its survival probability is

$$Pr(\text{block survives}) = (1 - p_t)^{l_i}. \quad (2.5)$$

The probability that at least one bit changes is

$$Pr(\text{block changes}) = 1 - (1 - e/P)^{l_i}. \quad (2.6)$$

These equations explain why short blocks tolerate variation better than long blocks. They also show that a hierarchy can contain levels that almost never match if its blocks are too long for the measured disagreement rate.

The published hierarchy grows the block length by a factor w [2]:

$$l_i = w l_{i-1}. \quad (2.7)$$

For hierarchy depth d , the total number of filters is

$$N = \sum_{i=0}^d \left\lceil \frac{P}{l_i} \right\rceil. \quad (2.8)$$

The ceiling means that a remainder block still receives a filter. This detail matters in the corrected implementation used in this thesis.

The paper reports a design example with $P = 4,900$, an average disagreement rate near 0.16, a low-level block length near 32, a growth factor near 13, and depth 2 [2]. This produces 32-bit blocks, 416-bit blocks, and a complete-template block. Full coverage gives 154 low-level blocks, 12 middle blocks, and one top block, for 167 filters. The published artifact reports 165 filters because final remainder blocks were omitted. Pipeline 1 uses 167 filters so that all template bits are covered. Figure 2.2 summarizes these three levels and their block lengths.

For a query q , let $I_i(q)$ be one when block filter i matches and zero otherwise. The HBF score is

$$s(q) = \sum_{i=1}^N I_i(q). \quad (2.9)$$

The query is accepted when $s(q)$ reaches a threshold selected with development identities. The score-and-threshold view is used for all retained pipelines in this thesis.

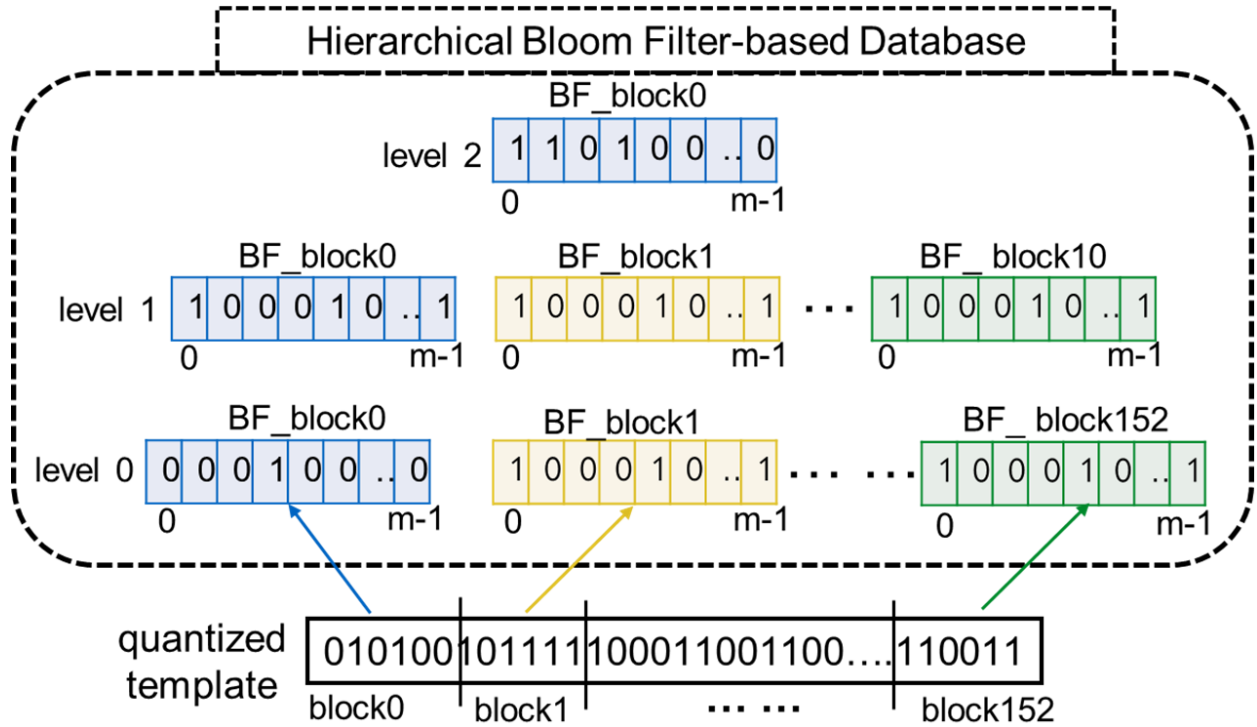


Figure 2.2. Hierarchical Bloom-filter organization. Source: Shomaji et al. [2, Fig. 3]. © 2019 IEEE.

2.6 Discrete Cosine Transform

The Discrete Cosine Transform (DCT) is a common transformation method for representing a signal using cosine patterns at different frequencies. Ahmed, Natarajan, and Rao introduced the DCT for digital signal processing and pattern-recognition applications [5]. When applied to an image patch, the 2-D DCT converts spatial pixel values into spatial-frequency coefficients.

For an $M \times N$ patch $I(x, y)$, the orthonormal two-dimensional DCT is

$$C(u, v) = \alpha_M(u)\alpha_N(v) \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} I(x, y) \cos \left[\frac{\pi(2x+1)u}{2M} \right] \cos \left[\frac{\pi(2y+1)v}{2N} \right], \quad (2.10)$$

where u and v are frequency indices. The normalization term is $1/\sqrt{M}$ when $u=0$ and $\sqrt{2/M}$ otherwise; $\alpha_N(v)$ is defined in the same way for N .

The coefficient at $(0,0)$ is the direct-current, DC coefficient. It mainly describes the average intensity of the patch. Low-frequency alternating-current coefficients describe broad intensity changes. Higher-frequency coefficients describe faster changes and fine detail. Chapter 5 preserves the existing low-frequency AC ordering and directly stores its first 12 binarized entries at every retained landmark.

2.7 ArcFace Spatial Features and PCA

Deep face-recognition networks learn representations in which images of the same identity are encouraged to remain close while different identities are separated. ArcFace improves this separation by introducing an additive angular margin during training [6].

Most face-recognition applications use the final whole-face embedding produced by the network. Pipeline 3 instead uses a spatial feature map. Local descriptors are sampled from this feature map at facial-landmark locations and then used with the same Landmark-Indexed Bloom Filter organization introduced in Pipeline 2.

The sampled ArcFace descriptors are high-dimensional real-valued vectors, so another step is required before they can fit into the compact binary representation used by the HBF. Pipeline 3 uses Principal Component Analysis (PCA) for this purpose. PCA provides a fixed linear bridge from the 512-dimensional descriptor to the 12-value regional budget: it retains the directions with the largest variation in the fit data instead of selecting arbitrary feature channels. This also reduces redundant correlated dimensions before sign binarization. Whitening then scales the retained directions so that no one high-variance component dominates the bit decisions [16]. In Pipeline 3, PCA whitening and a seeded rotation are fitted once and folded into one 512×12 matrix. Each sampled 512-dimensional descriptor can then be mapped directly to 12 values before binary conversion.

2.8 Bloom-Filter Template Protection

Biometric-template protection involves more than replacing a template with a hash. Recognition performance, recovery of biometric information, replacement following compromise, and linkage between different applications are all relevant properties [3].

The published implementation first applies SHA-256 and then uses an FNV-based operation to map the result into the Bloom-filter address range [2]. Its filter bank records positions produced from hashed blocks instead of storing the face image directly. This limits direct exposure of the image, although occupancy patterns, repeated blocks, and shared structure may still reveal information about the Bloom-filter representation.

The scope of this thesis is limited to recognition and efficiency. It does not provide a formal proof of irreversibility, unlinkability, or renewability. In this work, the term privacy-preserving refers to the operational property that the membership bank does not store raw face images and

does not directly return an identity. It should not be interpreted as evidence that every possible attack against the representation has been evaluated.

2.9 Contributions

Bloom introduced approximate membership testing as a space-time trade-off [1]. Shomaji et al. later applied a hierarchical version of Bloom filters to noisy biometric blocks [2]. Their reported storage result is from the published HBF study and is not an experimental result of this thesis. Shomaji et al. leave the choice of stronger HBF input features as an open question [2]. This question provides the starting point for the work presented here. In the published HBF, an aligned face becomes a single sequence of pixels binarized against a population mean. The hierarchy separates that sequence by block length rather than by facial anatomy. As a result, every image region initially uses the same type of feature and occupies the same structural role. This leaves two related questions open.

1. What facial information should be stored in the binary blocks?
2. How should those blocks be organized when different facial regions may not contribute equally to recognition?

This thesis studies these questions using **Landmark-Indexed Bloom Filters**. The published pixel-based HBF is first reproduced and diagnosed as Pipeline 1. The diagnostic results are then used to develop compact local representations centered at repeatable facial locations. Pipeline 2 uses local DCT features, while Pipeline 3 keeps the same region-indexed organization and replaces the local representation with identity-trained ArcFace spatial features.

Chapter 3: Datasets, Experimental Protocol, Preprocessing, and Evaluation

Metrics

3.1 Datasets

Five face datasets are used in this study: Faces94, Faces95, Faces96, FEI, and FERET. Together, they provide repeated face images with differences in expression, lighting, background, scale, and pose. This variation is useful for evaluating whether the membership representation can still recognize an enrolled person when the query image is different from the image used during enrollment.

Table 3.1 lists the dataset copies used for the experiments. The values correspond to the local data available after the datasets were organized and processed for this study.

Dataset	Identities	Images
Faces94	152	3,040
Faces95	72	1,440
Faces96	151	3,016
FEI	200	400
FERET	923	1,846

Table 3.1. Active dataset inventory.

3.1.1 Faces94, Faces95, and Faces96

Faces94, Faces95, and Faces96 belong to the University of Essex face collection created by Libor Spacek. The collection contains color face images, with its main subsets having different levels of acquisition difficulty. Faces96 contains greater background and scale variation than the earlier subsets [7].

Each retained identity has approximately 20 photographs. Faces94 is the most controlled of the three datasets. Faces95 introduces more background and lighting variation, while Faces96 contains stronger changes in background and scale. The counts in Table 3.1 correspond to the verified local copies used in the experiments [7].

The multiple photographs available for each identity also allow several images of the same person to be enrolled while keeping other images for genuine queries.

3.1.2 FEI

The complete FEI face database contains 14 color photographs for each of 200 individuals. A manually aligned frontal subset is also provided with two frontal photographs per identity, one with a neutral expression and one with a smiling expression [8]. The experiments in this study use this 400-image frontal subset.

Unlike the Essex datasets, only one FEI photograph per enrolled identity is used for enrollment. The second photograph is used as the genuine query. This gives the system only one enrolled appearance of each person and provides a more difficult condition when the expression changes between enrollment and query.

3.1.3 FERET

The FERET program developed a large face database and evaluation procedures for face-recognition research. The complete database contains 14,126 images of 1,199 individuals collected across several sessions [9]. The subset used here contains 923 identities and 1,846 images.

For each retained identity, one frontal photograph is used for enrollment, and one frontal photograph is used for evaluation. FERET therefore provides another one-image enrollment

setting, like FEI. It also gives an additional acquisition source beyond the Essex and FEI datasets. The same FERET subset and protocol are used for all retained pipeline comparisons.

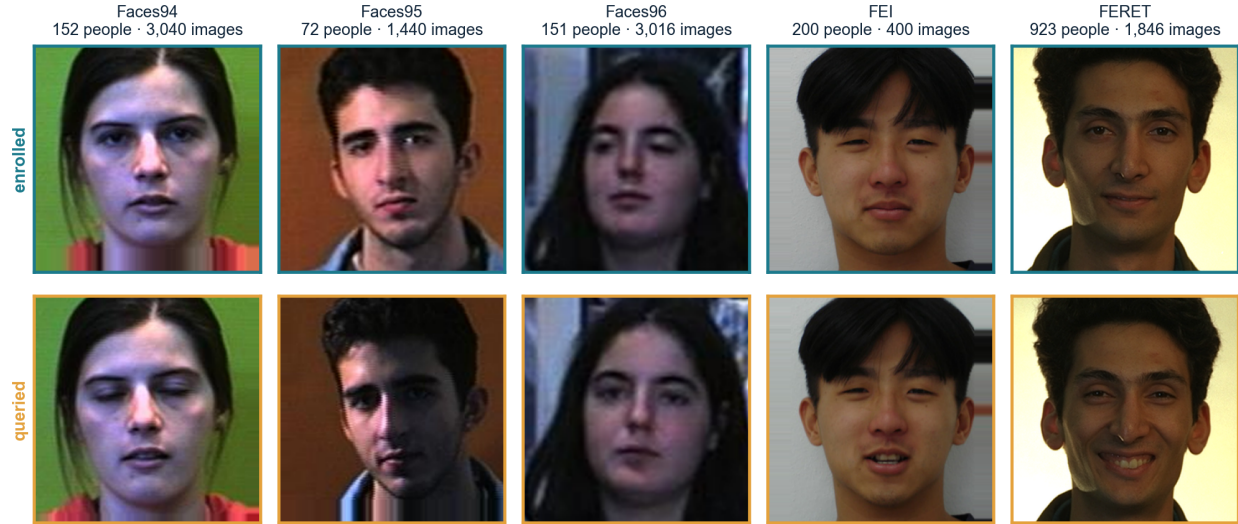


Figure 3.1. Representative images from the five datasets used in this thesis [7], [8], [9]. Images are shown only to explain the experimental data and are not stored in the HBF banks.

3.2 Identity-Disjoint Fit, Development, and Evaluation Partitions

The 1,498 identities are partitioned before any representation parameters or operating thresholds are fitted. The fit partition contains 450 identities (30%), the development partition contains 375 identities (25%), and the evaluation partition contains 673 identities (45%). No identity appears in more than one partition.

The fit identities are used to estimate the mean face and all learned representation parameters. The development identities are used to select the landmark subset, descriptor and code settings, and dataset-specific membership thresholds. The evaluation identities remain untouched until these choices are frozen and are used only to produce the final reported scores.

The same identity assignments and enrollment, genuine-query, and impostor-query roles are reused for all three pipelines. This keeps the comparisons identity-disjoint and ensures that differences between pipeline results do not come from fitting or evaluating different people.

3.3 Enrollment and Query Protocol

Each dataset bank contains only photographs belonging to identities selected for enrollment.

Identities used as impostors are never inserted into the bank.

For Faces94, Faces95, and Faces96, successfully processed photographs belonging to an enrolled identity are divided deterministically. The first portion is used for enrollment, giving several enrolled photographs where available. The remaining images of that identity are used as genuine queries.

FEI and FERET follow a different enrollment condition because the accepted subsets provide only two photographs per identity. One photograph is enrolled and the other is used as the genuine query. In every dataset, impostor queries are taken from held-out identities that do not appear in the enrolled bank.

3.4 Common Face Preprocessing

All three pipelines begin with the same dlib-based face localization and landmark alignment. Processing continues only when exactly one HOG-based frontal face is detected [21]. The 68-point landmark predictor is then applied. Starting from a mean shape inside the detected face region, the predictor refines the landmark locations through a cascade of sparse shape-indexed pixel-intensity tests [4].

The left-eye center is calculated using landmarks 36 through 41, while landmarks 42 through 47 are used for the right-eye center. The mouth center is calculated from the two mouth-corner

landmarks. These locations define an affine transformation that maps the face to fixed eye and mouth positions in a 160×160 aligned image.

Images that cannot be read, contain an incorrect number of detected faces, or fail during transformation are recorded as failures. A failed image is not replaced with an arbitrary template.

The retained pipelines use the same set of successfully processed images.

Pipeline 1 performs several additional steps after this common alignment. An elliptical mask removes much of the hair, ears, neck, and background while retaining the main internal facial area. Histogram equalization is applied to reduce large global lighting differences. The image is then resized to 70×70 pixels and flattened into 4,900 values.

3.5 Development Choices and Frozen Decisions

Evaluation identities are excluded from all fitted statistics and development decisions. This includes the Pipeline 1 mean face as well as the feature thresholds, transformations, and landmark selection used by the later pipelines.

Membership thresholds are selected using development identities. Once selected, each threshold is frozen and applied to the evaluation identities without further adjustment. Threshold selection is performed separately for each dataset and pipeline because their score distributions are different.

This protocol prevents final evaluation results from being tuned after the evaluation labels have been observed. As a result, each reported accuracy value corresponds to a threshold chosen before that final evaluation.

3.6 Evaluation Metrics

The reporting scope of this thesis is limited to AUC, accuracy, and storage. Table 3.2 summarizes the purpose of each measure.

Measure	Purpose	Reporting level
AUC	Measures how well genuine queries rank above impostor queries across thresholds	Each dataset and equal-dataset macro AUC
Accuracy	Measures correct decisions at one frozen threshold	Each dataset
Storage	Measures template and HBF-bank size	Each pipeline and dataset where applicable

Table 3.2. Evaluation measures.

3.6.1 Area Under the ROC Curve

A receiver operating characteristic (ROC) curve evaluates the trade-off between genuine and impostor acceptance as the decision threshold changes. AUC summarizes this ranking behavior across thresholds [10]. An AUC of 1.0 represents perfect ranking, while an AUC close to 0.5 indicates approximately chance-level ranking.

Per-dataset AUC is the main recognition measure used in this study. Let AUC_d be the AUC for dataset d and let $D=5$ be the number of datasets. The equal-dataset macro AUC is

$$\text{Macro AUC} = \frac{1}{D} \sum_{d=1}^D AUC_d. \quad (3.1)$$

Each dataset receives equal weight, regardless of the number of queries it contains. No other aggregate recognition score is used.

3.6.2 Accuracy

Accuracy is calculated using the membership threshold selected on the development identities and frozen before evaluation. Let TP represent accepted genuine queries, TN rejected impostor queries, FP accepted impostor queries, and FN rejected genuine queries. Accuracy is

$$\text{Accuracy} = \frac{TP+TN}{TP+TN+FP+FN}. \quad (3.2)$$

Accuracy is reported for each dataset using its frozen dataset-specific threshold. For the cross-pipeline summary in Chapter 6, equal-dataset macro accuracy is also reported as the arithmetic mean of the five per-dataset accuracies. This summary gives each dataset equal weight and does not replace the per-dataset results.

3.6.3 Storage

Two types of storage are considered: the binary template produced for one image and the complete HBF bank.

$$\text{Template bytes} = \left\lceil \frac{P}{8} \right\rceil. \quad (3.3)$$

HBF-bank storage measures all Bloom-filter arrays for one dataset bank. If a bank contains N filters of m bits, its storage is

$$\text{HBF-bank bits} = N \times m. \quad (3.4)$$

The packed representation is rounded to complete bytes. Template storage and HBF-bank storage are reported separately because they represent different parts of the system.

3.7 Reproducibility

The experimental seed is fixed at 42. Dataset assignments, preprocessing geometry, template lengths, filter parameters, and model selections are recorded for the retained experiments. SHA-

256 checksums are also recorded for the landmark predictor and ArcFace checkpoint. Before inclusion in the thesis, the primary AUC and accuracy results were independently recomputed from the saved prediction outputs.

The saved dataset records include the false-positive target, filter count, enrolled population, filter size, hash count, frozen threshold, and storage. This allows the reported results to be traced back to the configuration used to produce them.

The repeated timing run records macOS 15.5 on Apple Silicon and Python 3.12.5. Timing remains hardware- and software-dependent, and the five repetitions quantify backend-query variability only; no complete image-to-decision claim is made.

Chapter 4: Baseline HBF and Diagnostic Analysis

4.1 Purpose of the Baseline

Before introducing the region-aware pipelines, the original Hierarchical Bloom Filter design was implemented and evaluated as Pipeline 1: Baseline HBF. The implementation follows the HBF architecture described by Shomaji et al. [2], including the global mean-thresholded pixel representation and the low-, middle-, and complete-template hierarchy. The purpose of this baseline is not only to provide a result for comparison. It also provides a starting point for examining where the original HBF works and where the global hierarchical pixel representation begins to lose useful identity information under the experimental protocol used in this study. Before comparing the proposed pipelines with the baseline, the implementation was checked to make sure that the HBF enrollment and query process was working correctly. When an exact template that had already been enrolled was queried, all expected filters were activated. The hierarchy was also checked to make sure that all 4,900 input bits were covered. These checks are useful because poor performance can then be studied as a limitation of the representation or hierarchy rather than as an error in the Bloom-filter implementation.

4.2 Baseline HBF Architecture

This study addresses the Baseline HBF [2] as Pipeline 1.

Pipeline 1 consists of five main stages: face alignment, global pixel preparation, mean-based binary conversion, hierarchical block construction, and Bloom-filter membership scoring.

4.2.1 Face Alignment and Pixel Preparation

The preprocessing begins by detecting one face and estimating 68 facial landmarks. The detected face is aligned using fixed eye and mouth positions to produce a 160×160 grayscale image. An elliptical mask is then applied to retain the internal facial region. Histogram equalization is used to reduce larger lighting differences, after which the masked image is resized to 70×70 pixels. The 70×70 image is flattened in row-major order. Since every pixel becomes one feature, the resulting feature vector contains 4900 values. The spatial order is retained in the vector.

$$70 \times 70 = 4,900 \quad (4.1)$$

4.2.2 Mean-Thresholded Binary Template

The baseline method estimates a population mean for each feature position and compares every aligned face with that reference [2]. Pipeline 1 follows the same idea without using evaluation identities to fit the mean. Let x_j be pixel j of an aligned face and let μ_j be the fitted mean at that location. The binary template bit is

$$b_j = \begin{cases} 1, & x_j > \mu_j, \\ 0, & x_j \leq \mu_j. \end{cases} \quad (4.2)$$

Applying (4.2) to every pixel creates a 4,900-bit template. Its packed size is

$$\left\lceil \frac{4,900}{8} \right\rceil = 613 \text{ bytes.} \quad (4.3)$$

This conversion reduces storage because an eight-bit grayscale value no longer needs to be stored for every pixel. However, binarization alone does not mean that every retained bit contains useful identity information. This issue is examined later in Section 4.7.

4.2.3 Hierarchy Construction

Once the 4,900-bit template is created, it is divided into blocks at three hierarchy levels. The lowest level contains blocks of up to 32 bits. The middle level contains blocks of up to 416 bits, and the top level consists of one complete 4,900-bit block.

The published implementation reports 165 filters because the final partial blocks at the lowest and middle levels were omitted [2]. In Pipeline 1, these remaining bits are also included so that the complete template is represented. The hierarchy is therefore

$$\left\lceil \frac{4,900}{32} \right\rceil = 154, \left\lceil \frac{4,900}{416} \right\rceil = 12, 1 \text{ top block.} \quad (4.4)$$

The complete baseline uses

$$154 + 12 + 1 = 167 \quad (4.5)$$

Bloom filters. The remainder blocks are shorter than 32 or 416 bits, but they are still hashed as separate position-specific blocks.

4.2.4 Hashing and Bloom-Filter Bank

The baseline reported by Shomaji et al. uses SHA-256 for the initial digest and an FNV operation for Bloom-filter addressing [2]. The audited implementation used here keeps the same general two-stage purpose while using a keyed construction. HMAC-SHA-256 is first applied to a domain containing the hierarchy level, block name, block length, and packed block bits [18], [19]. FNV-1a is then used to fold the result into the valid Bloom-filter address range [20]. A counter is used to generate k addresses. FNV-1a is used only for address folding and is not considered a cryptographic hash.

Including the hierarchy level and block name in the hashed domain is necessary because the same binary block may occur at different locations. Without this information, identical bit strings

from two different block positions could be treated as the same HBF item. The secret key also allows banks from different deployments to be separated. This keyed construction is an implementation choice and should not be interpreted as a formal template-protection analysis. For the unified evaluation, the target false-positive probability for an individual Bloom filter is set to 0.01. The required filter length is calculated separately for each dataset using the number of enrolled photographs and (2.1). The number of hash addresses is selected using (2.2). The accepted Pipeline 1 banks use $k=7$.

4.2.5 Query Score

A query image passes through the same alignment and binary conversion steps and produces a 4,900-bit template. This template is divided into the same 167 position-specific blocks. Each block receives a value of one when all of its required Bloom-filter addresses are set and zero otherwise. The final HBF score is the sum of the matched blocks, following (2.9). The decision threshold is selected using development identities and then frozen before evaluation.

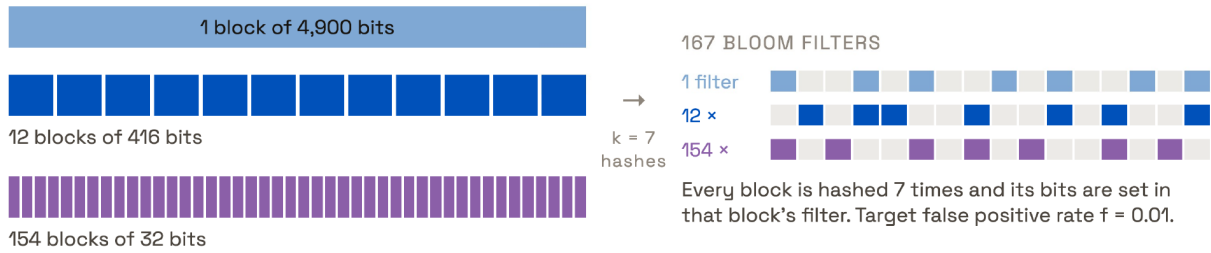


Figure 4.1. Pipeline 1 architecture. The face becomes a 4,900-bit global pixel template before hierarchical enrollment. The processing flow is based on [2] and implemented by this study.

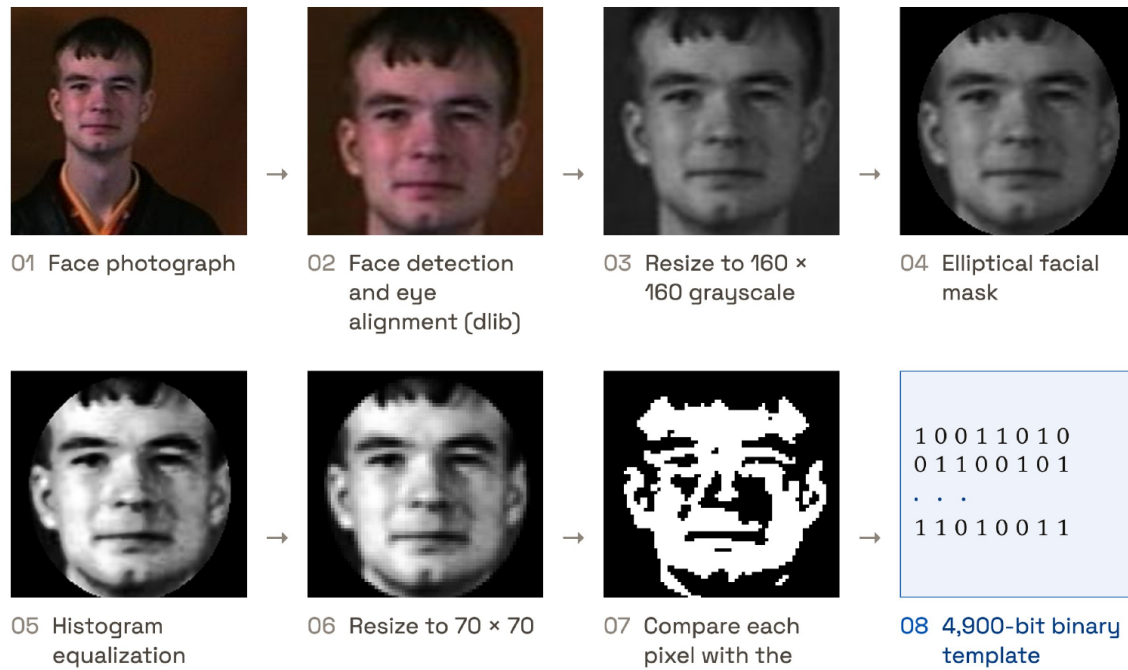


Figure 4.2. Pipeline 1 processing stages using an image processed by the thesis implementation.

4.3 Pipeline 1 Results

Table 4.1 presents the Pipeline 1 results under the primary evaluation protocol. For each dataset, the decision threshold was selected using development identities and then frozen before evaluation.

Dataset	AUC	Accuracy	Correct / total	HBf-bank storage (bytes)
Faces94	0.9140	0.8315	696 / 837	101,662
Faces95	0.7064	0.6438	253 / 393	46,030
Faces96	0.6498	0.6655	553 / 831	101,056
FEI	0.5672	0.6667	60 / 90	13,611
FERET	0.5395	0.5680	234 / 412	61,832

Table 4.1. Pipeline 1 evaluation results.

The equal-dataset macro AUC is

$$\frac{0.9140+0.7064+0.6498+0.5672+0.5395}{5} = 0.6754. \quad (4.6)$$

There is a substantial difference in performance across the five datasets. Faces94 reaches an AUC of 0.9140, showing that the pixel representation can work under relatively controlled conditions with several enrollment photographs. Faces95 and Faces96 show more overlap between genuine and impostor scores. FEI and FERET remain close to chance in AUC under one-image enrollment.

The five HBF banks require a total of

$$101,662 + 46,030 + 101,056 + 13,611 + 61,832 = 324,191 \text{ bytes.} \quad (4.7)$$

4.4 Diagnostic Experiments: Overview

The baseline results show that Pipeline 1 has limited performance on several datasets, but the results alone do not identify the cause. The next set of experiments was used to examine the HBF more closely before changing its architecture. Four questions are considered: how much each hierarchy level contributes, whether Bloom-filter size is limiting performance, what information is actually present in the 4,900-bit binary template, and whether previous work supports giving different facial regions different importance.

These diagnostic experiments are used to study the mechanism behind the baseline behavior.

Their values are not replacements for the primary-protocol AUC and accuracy results reported in Table 4.1. Instead, each experiment isolates or changes one part of the system so that its effect can be examined separately.

4.5 Diagnostic Experiment 1: Hierarchy-Level Contribution

The first experiment examines whether the lowest, middle, and top hierarchy levels all contribute useful membership information. Rather than combining all 167 Bloom-filter outputs into one score, the levels are scored separately.

The measured bit-disagreement ratio for the baseline is approximately $e/P = 0.1165$. Equation (2.5) gives the independent-bit approximation for the survival probability of an exact block. For the three hierarchy levels, the predicted probabilities are 0.01899 for a 32-bit block, 4.19×10^{-23} for a 416-bit block, and 2.58×10^{-264} for the complete 4,900-bit block. These values are model-based predictions under the independence assumption, not measured Bloom-filter firing rates.

The measured behavior follows the same general pattern. Genuine queries activate approximately 42.7% of the lowest-level filters, 8.7% of the middle-level filters, and 1.1% of the top filter. At the upper levels, the impostor firing rates are also close to the genuine firing rates. When each level is evaluated separately, the lowest level gives an AUC of 0.639, the middle level gives 0.544, and the top level gives 0.499.

Table 4.2 summarizes the hierarchy structure, model-based survival predictions, observed role, and level-specific discrimination.

Level	Blocks	Bits per block	Predicted survival	Observed role	AUC alone
Lowest	154	32	0.01899	Dominant	0.639
Middle	12	416	4.19×10^{-23}	Minimal	0.544
Top	1	4,900	2.58×10^{-264}	None	0.499

Table 4.2. Hierarchy-level survival and discrimination under the diagnostic evaluation.

An AUC of 0.499 places the top level essentially at chance. The middle level contains only a small amount of ranking information, while most of the useful signal comes from the 154 low-

level filters. Under this evaluation, the upper hierarchy levels consume HBF-bank storage while contributing little useful separation.

4.6 Diagnostic Experiment 2: Bloom-Filter Sizing

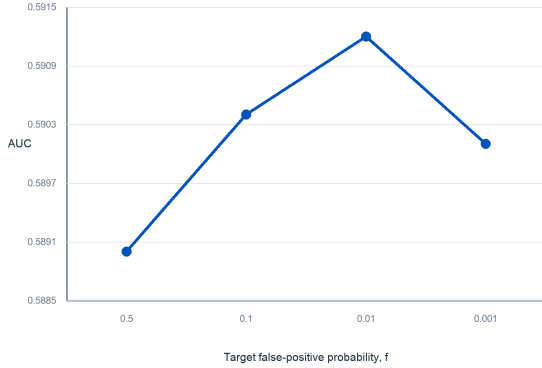
The second experiment tests whether Pipeline 1 can be improved simply by changing the Bloom-filter false-positive target. The target is varied over a wide range, and an ideal collision-free membership check is also evaluated. This gives a useful reference for separating problems caused by Bloom-filter collisions from problems already present in the input representation.

Experiment 2A varies the target false-positive probability f across 0.5, 0.10, 0.01, and 0.001. The corresponding diagnostic AUC values are 0.5890, 0.5904, 0.5912, and 0.5901. The total movement across the sweep is therefore 0.0022, and the selected $f=0.01$ setting gives the highest value in this experiment.

Experiment 2B compares the selected Bloom-filter setting with a collision-free exact-set membership check. The $f=0.01$ Bloom-filter result is 0.5912, while the exact-set result is 0.5907, a difference of 0.0005. Removing Bloom-filter collisions therefore does not improve the diagnostic AUC.

The small change across the sweep indicates that Bloom-filter collisions are not the main source of the baseline limitation. A larger value of m can reduce collision probability, but it cannot separate genuine and impostor blocks when the block values themselves are already similar. Increasing the Bloom-filter size further would mainly increase storage without fixing the representation problem.

Experiment 2A: Bloom-filter target sweep



Experiment 2B: Exact-set reference

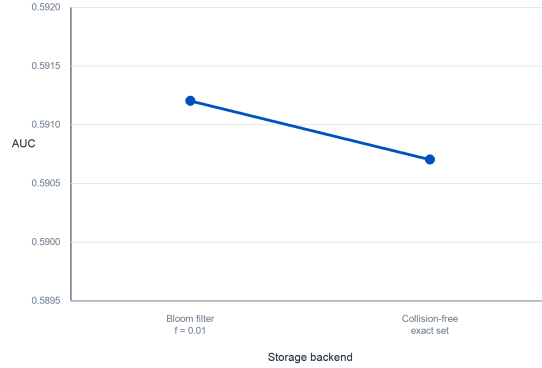


Figure 4.3. Bloom-filter sizing results for Experiment 2A and the exact-set reference in Experiment 2B. The numerical comparisons are reported in the accompanying text.

4.7 Diagnostic Experiment 3: Information Inside the Binary Template

The third experiment looks directly at the 4,900-bit mean-thresholded pixel template. Since every aligned image is compared with a fitted mean face, some of the resulting bits may mainly describe facial structure that is common across individuals rather than differences between identities.

The analysis shows that 1,635 of the 4,900 bit positions are identical for every enrolled person in the diagnostic gallery. The combination of alignment, the elliptical mask, and the global mean threshold creates a broad face-shaped pattern that is shared across identities. In addition, 12 complete 32-bit blocks are constant across the gallery.

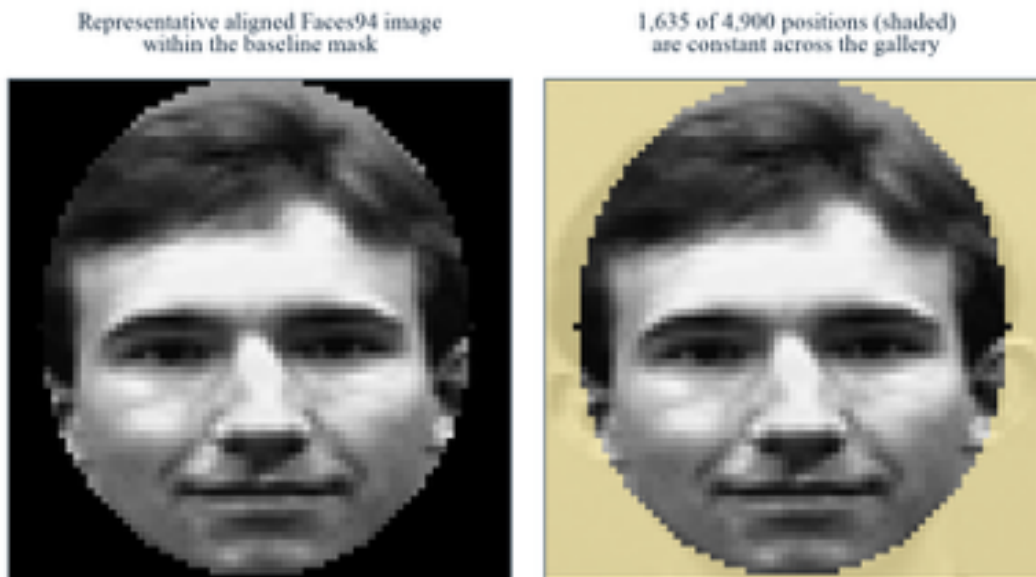


Figure 4.4. Information inside the baseline template. A representative aligned Faces94 image is shown; light-yellow positions are constant across the diagnostic gallery [7].

These 12 blocks are ordinary position-specific blocks. They are not special filters added to represent the elliptical mask. Their masked and mean-thresholded inputs simply happen to be identical for every enrolled identity. Since the same block value can occur for everyone, such a block does not provide identity separation.

This can also be seen in the firing rates of the lowest hierarchy level. Genuine queries activate 42.7% of the low-level filters, while impostor queries activate 37.9%. The ratio between these values is only about 1.1. When the Bloom-filter layer is removed and the exact block values are compared directly, the AUC changes by only 0.0013 in the retained diagnostic experiment.

This result is consistent with the Bloom-sizing experiment. Accidental Bloom-filter collisions are not the main issue. Many of the global pixel blocks are already shared across different people before they are inserted into the Bloom filters. The baseline records whether a pixel is brighter or darker than the corresponding position in an average face, and many of these comparisons capture common facial structure rather than identity-specific information.

4.8 Diagnostic Experiment 4: Literature Review of Facial-Region Importance

The previous diagnostic experiments point toward the representation itself, but they do not indicate which parts of the face should receive more attention. Previous studies provide some motivation for testing facial regions separately in Pipeline 2.

Sadr, Jarudi, and Sinha studied the role of eyebrows in familiar-face recognition and found that removing the eyebrows caused a larger reduction in recognition than removing the eyes [22].

Vinette, Gosselin, and Schyns reported that information around the eyes becomes diagnostic quickly during face identification [23].

Other studies have reported similar importance for the eye and periocular regions. Royer et al. found that greater reliance on the eye region was associated with better face-recognition ability [24]. Periocular-biometric research has also shown that the region surrounding the eyes can support recognition [25]. Peterson and Eckstein found that an effective viewing location for several face-recognition tasks lies just below the eyes [26].

These findings provide motivation for a periocular-first hypothesis, but they do not define the optimal HBF landmark configuration for the datasets, DCT frontend, and membership-verification task used in this study. They also do not establish that regions such as the mouth contain no useful information.

For this reason, the literature is used only to motivate the regional analysis. The actual landmark utility, regional ablation, and landmark-count selection used by Pipeline 2 are measured on development identities and are frozen before final evaluation.

4.9 Combined Baseline Diagnosis

Table 4.3 summarizes the main findings from the diagnostic experiments.

Diagnostic question	Main evidence	Design meaning
Do all hierarchy levels help?	The top level has AUC 0.499; the middle level adds little signal	Do not spend storage on long levels that rarely survive
Is Bloom-filter size the main limit?	AUC changes by only 0.0022; a perfect filter reaches only 0.5907 in the diagnostic setting	More backend storage will not repair the representation
Do the bits identify people well?	1,635 bits and 12 blocks are constant; genuine and impostor firing rates are close	Transform local appearance before binary conversion
Which regions should Pipeline 2 test?	Prior work emphasizes eyebrows, eyes, and periocular structure	Measure landmark utility and ablation on development identities

Table 4.3. Summary of the baseline diagnosis.

Taken together, the experiments show that the published HBF backend can correctly store and query the binary blocks, while the global mean-thresholded pixel representation does not make efficient use of identity information under this protocol. The hierarchy also allocates storage to long blocks that rarely survive between photographs of the same person. At the lowest level, many blocks still represent facial structure that is shared by different identities.

The main limitation of the baseline is therefore associated with the representation and how that representation is organized. Once the block values have been produced, changing the Bloom-filter parameters has only a small effect on performance.

4.10 Requirements for a Region-Aware Design

The diagnostic results provide several requirements for the next pipeline. The upper hierarchy levels contribute little useful ranking information, so these levels should not automatically be retained. Filters should also correspond to repeatable facial locations rather than arbitrary positions in a flattened global pixel sequence.

The local appearance around each region should be transformed before binary conversion. This is intended to move the representation away from simple comparisons with common facial brightness structure. The results also suggest that using every facial location equally may not be necessary, so the next design should allow a smaller subset of useful regions to be retained.

At the same time, the basic HBF membership mechanism should remain simple. The final decision should still be based on a Bloom-filter match score and a development-selected threshold. Keeping this backend structure makes it possible to study whether improvements come from the representation and regional organization rather than from increasing the size or complexity of the membership backend.

Chapter 5: Landmark-Indexed Bloom Filters

Chapter 4 showed that the global pixel representation and its hierarchy are not well suited to localized facial evidence. Based on that diagnosis, this chapter develops Landmark-Indexed Bloom Filters (LBFs) as a common region-aware framework and evaluates two different descriptor frontends within it. DCT-LBF uses compact handcrafted frequency features, while Deep Spatial LBF uses identity-trained contextual features. Both use the same landmark-indexed backend and the same membership-testing interpretation.

5.1 Proposal of Landmark-Indexed Bloom Filters

The diagnostic experiments motivate a different way of organizing the Bloom-filter bank. In the original HBF, the face is represented as one global binary string and divided into hierarchical pixel blocks [2]. The results in Chapter 4 showed two problems with this organization: useful facial evidence is not distributed evenly across the face, and long blocks are unstable between photographs of the same person. This motivates storing smaller local representations at repeatable facial locations instead of combining large sections of one global binary template. Landmark-Indexed Bloom Filters (LBFs) are proposed for this purpose. Each retained facial landmark is assigned one compact local descriptor and binary code, and that code is stored and queried only through the Bloom filter associated with that landmark. The landmark address is therefore part of the membership item. The same code occurring at two different facial locations is not treated as equivalent.

This organization is shared by the two proposed variants. Pipeline 2: DCT-LBF uses a handcrafted low-frequency DCT code at each landmark. Pipeline 3: Deep Spatial LBF replaces that local descriptor with a frozen identity-trained representation while keeping the same

landmark-indexed backend.

Both variants follow the same LBF structure: 48 development-selected landmark addresses, one 12-bit code for each landmark, and one dedicated Bloom filter for each location. This gives a 576-bit, or 72-byte, packed query template. The 48 Bloom filters each produce a binary response, and the membership score is the number of filters that fire. Figure 5.1 shows this common backend. The descriptor used to produce each 12-bit code is the controlled difference between DCT-LBF and Deep Spatial LBF.

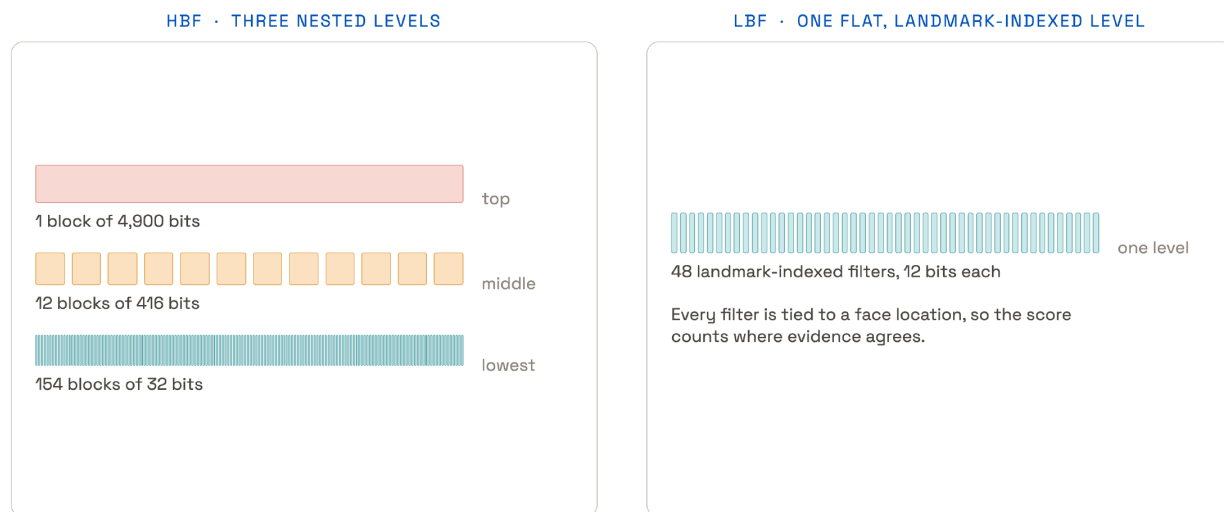


Figure 5.1. Common Landmark-Indexed Bloom Filter backend. Each retained landmark produces one 12-bit local code that is stored and queried only in its dedicated Bloom filter; the 48 filter responses form the membership score.

5.2 DCT-LBF

5.2.1 Motivation

The baseline diagnosis suggests that both the representation and its organization need to change. Pipeline 1 represents the entire face as a global pixel string even though different facial regions contain different amounts of identity information. It also retains middle and top hierarchy levels that rarely match for genuine queries.

DCT-LBF addresses these findings by working with local facial regions instead. Each retained landmark produces a compact DCT code that is stored in a Bloom filter dedicated to that anatomical location. The middle and complete-template hierarchy levels are removed, so each selected landmark contributes one short regional block rather than taking part in larger concatenated blocks.

The design has three main goals. The first is to associate each block with a meaningful and repeatable facial location. The second is to describe local appearance rather than simply recording whether global pixels fall above or below a mean face. The third is to keep both the binary template and the HBF bank compact enough for efficient membership testing.

5.2.2 Architecture and Methodology

Pipeline 2 begins with the common 160×160 aligned grayscale face. Before final evaluation, identity-disjoint development identities are used to select and freeze the 32×32 to 8×8 sampling geometry, the 12-bit code length, the 48 retained landmark locations, and the decision threshold. Figure 5.2 summarizes the resulting execution path.

DCT-LBF remains an HBF-based membership system. It does not perform nearest-neighbor identification and does not return a named identity. The important structural change is that Bloom filters are indexed by facial anatomy rather than by positions in a flattened global pixel sequence.

Figure 5.3 follows the processing stages of DCT-LBF using one representative face already shown in Figure 3.1. It is included as an illustration of the processing sequence rather than as an additional experiment.

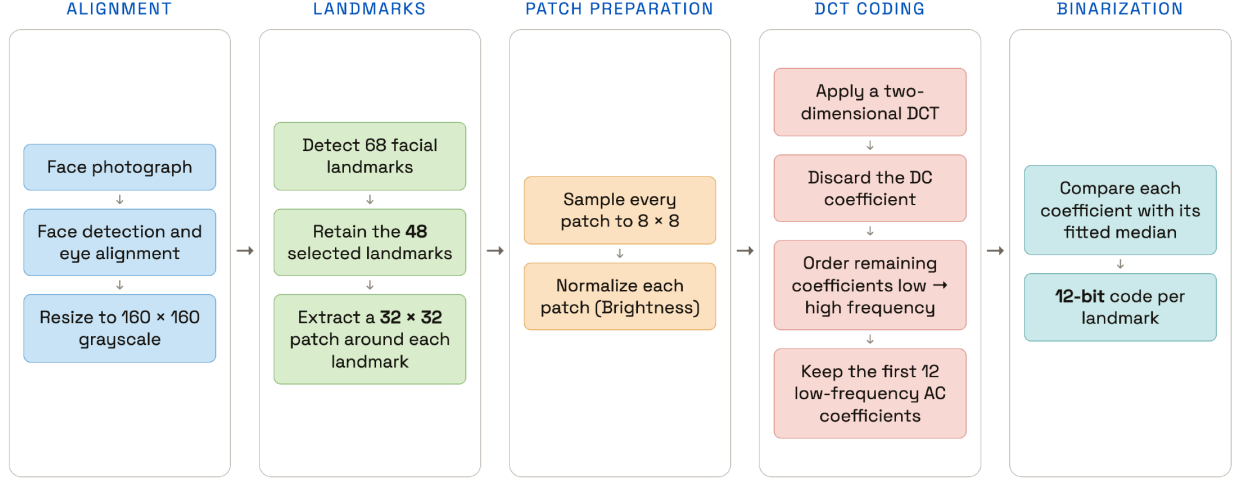


Figure 5.2. Pipeline 2 architecture. The development-frozen path converts each retained landmark region into one 12-bit DCT code and one region-indexed Bloom-filter query.

For a landmark with aligned coordinate (x_r, y_r) , Pipeline 2 defines a square region with a width of 32 pixels. Rather than cropping the 32×32 region and resizing it in a separate step, the implementation directly places an 8×8 sampling grid over the window. Grayscale values are read using bilinear interpolation. When a landmark, particularly along the jaw, lies close to the image boundary, reflection is used to supply values outside the image.

Let $I_r(a, b)$ be the sampled value at row a and column b of landmark region r . The patch is standardized as

$$\tilde{I}_r(a, b) = \frac{I_r(a, b) - \bar{I}_r}{\max(\sigma_r, 10^{-6})}, \quad (5.1)$$

where $\bar{I}_r$ and σ_r are the mean and standard deviation of the 8×8 patch. This standardization reduces differences caused by local brightness and contrast. The small constant prevents division by zero when a patch contains almost no variation.

The same 32×32 to 8×8 sampling geometry is used for every aligned face. This geometry was selected during development to keep the descriptor centered on a local anatomical region while

retaining enough surrounding structure to be useful. The landmark does not make two patches identical across photographs, but it gives the descriptor a repeatable location.

The standardized 8×8 patch is then transformed using the orthonormal two-dimensional DCT defined in (2.10) [5]. The transform produces 64 coefficients. The coefficient at (0,0), the DC coefficient, is discarded because it mainly represents average intensity, which has already been reduced through patch standardization. DCT has been used for compact facial appearance representation after geometric and illumination normalization [5], [27], as well as in local block-based face analysis [28].

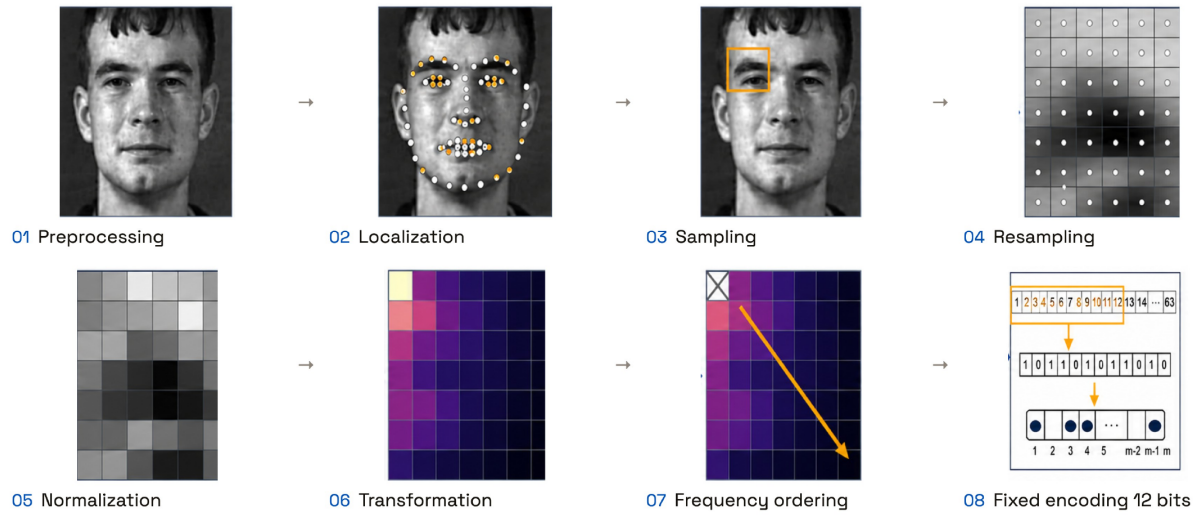


Figure 5.3. Pipeline 2 (DCT-LBF) processing phases on one representative photograph. The illustration traces alignment, retained landmarks, local sampling, DCT transformation, 12-bit regional encoding, and the common 48-filter LBF backend.

After removing the DC coefficient, the remaining AC coefficients are ordered by increasing $u+v$, with ties resolved using increasing u . Only the first 12 low-frequency AC positions are binarized and stored. The descriptor path is therefore 64 DCT coefficients to 12 stored bits, using the same fixed coefficient prefix at every retained landmark.

This changes what is being compared with the baseline. Pipeline 1 compares individual pixels with a global mean face, while DCT-LBF compares local spatial-frequency patterns with fitted coefficient references. Small movements can alter several individual pixels while still preserving part of the lower-frequency structure within a local region.

5.2.3 Development Choices

The DCT coefficients are converted to binary values without using evaluation identities. For dataset d , landmark r , and coefficient j , the fit identities provide a median $M_{d,r,j}$. The binary value is

$$b_{d,r,j} = \begin{cases} 1, & C_{d,r,j} > M_{d,r,j}, \\ 0, & C_{d,r,j} \leq M_{d,r,j}. \end{cases} \quad (5.2)$$

This produces a binary code at each of the 68 candidate landmarks. The fitted median provides a simple reference for binarization and also helps avoid codes dominated almost entirely by zeros or ones in the fit data.

With the coefficient order and fit-set medians fixed, a development sweep was used to compare code lengths of 4, 6, 8, 12, 16, 24, 32, and 48 bits per landmark. Every candidate used the same identity-disjoint fit, development, and evaluation partitions. Evaluation identities were not used to select the code length or decision threshold. The 12-bit configuration gives the highest development macro AUC at 0.9029, narrowly ahead of the 16-bit configuration at 0.8974.

Table 5.1 lists the choices made using development identities and frozen before final evaluation.

Choice	Candidates or conditions	Development evidence	Frozen configuration
Sampling geometry	Local window and resampled-grid dimensions	32×32 around the landmark, sampled to 8×8 , preserved regional structure while remaining local	$32 \times 32 \rightarrow 8 \times 8$
Bit length	4, 6, 8, 12, 16, 24, 32, and 48 bits	Development macro AUC peaked at 0.9029 for 12 bits; 16 bits reached 0.8974	12 bits per landmark
Landmark count	Top 24, top 31, top 48, and all 68 candidates	Top 48 reached 0.801 versus 0.802 for all 68, with 20 fewer region filters	48 landmarks
Decision threshold	Dataset-specific integer match-count thresholds	Selected on development identities after the representation was frozen	One frozen τ per dataset

Table 5.1. Development-frozen Pipeline 2 selections.

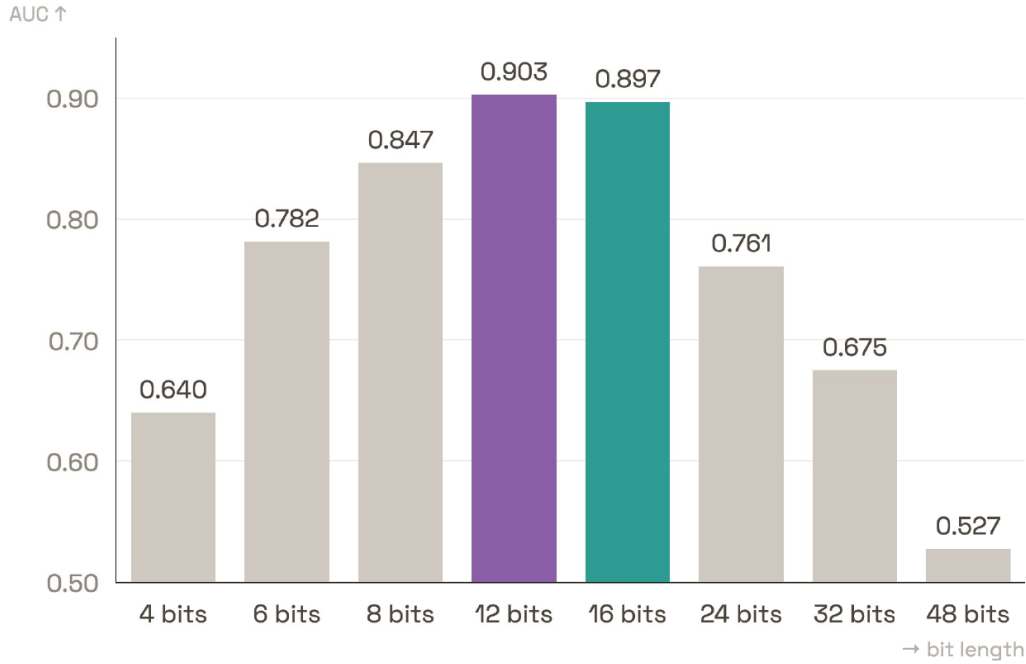


Figure 5.4. Pipeline 2 development-stage bit-length sweep. 12 bits is selected and frozen because it gives the highest equal-dataset development macro AUC.

The bit-length sweep shows two different failure cases. Very short codes of 4 to 8 bits are compact, but unrelated local appearances are more likely to produce the same code. At the other end, 24- to 48-bit codes are more specific but are less repeatable between photographs of the same person. The 12-bit code provides the best development result while keeping the full Pipeline 2 template at $48 \times 12 = 576$ bits, or 72 packed bytes. The selected length is applied globally, so every retained landmark uses the same first-12 low-frequency AC prefix.

A second development-only sweep was used to select the number of landmarks. The 68 candidate locations were first ranked by genuine firing rate minus impostor firing rate. The ranked list was then reduced to the top 24, top 31, and top 48 landmarks and compared with the full set of 68. The top-48 set reaches a development macro AUC of 0.801, compared with 0.802 for all 68 landmarks. Performance falls to 0.782 for the top 31 and 0.762 for the top 24. The 48-

landmark configuration was selected as the compact knee because it removes 20 region filters, or 29.4%, for a development AUC difference of only 0.001.

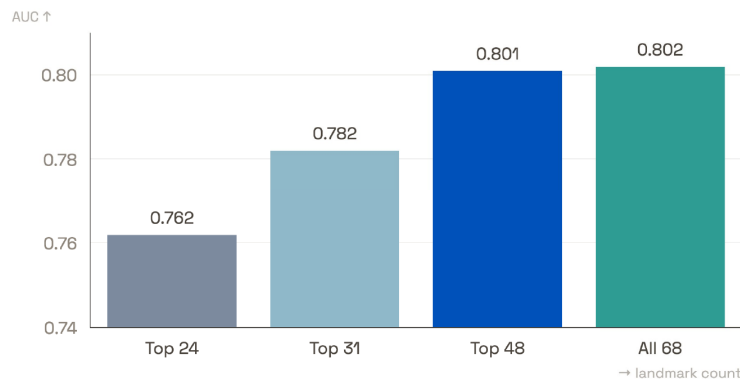


Figure 5.5. Pipeline 2 landmark-count development sweep. The usefulness-ranked top 48 retains nearly all of the all-68 development AUC while eliminating 20 region filters.

The final 48-location set contains all 10 brow landmarks, 11 of the 12 eye landmarks, and 9 of the 20 mouth landmarks. The remaining 18 locations consist of 11 jaw landmarks and 7 nose or nose-tip landmarks. These counts were produced by the usefulness ranking and were not imposed as predefined anatomical quotas. The exact 48-location list was frozen before evaluation and is reused unchanged in Pipeline 3.

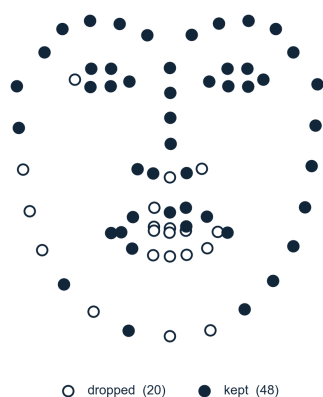


Figure 5.6. The 48 facial landmarks retained by the region-aware pipelines. Selection uses development evidence and remains frozen during final evaluation.

The selected landmark set remains fixed during evaluation and is shared by both region-aware pipelines.

The literature reviewed earlier motivates a periocular hypothesis, which is tested during Pipeline 2 development [22]–[26] without using evaluation identities. Each of the 68 landmark-centered candidates is scored using genuine firing rate minus impostor firing rate. The average regional scores are 0.131 for brows, 0.101 for eyes, 0.088 for jaw, 0.081 for nose, and 0.059 for mouth. A regional ablation experiment gives further evidence that the regions do not contribute equally. Removing the brow region reduces AUC by 0.0326, removing the eyes reduces it by 0.0171, and removing the mouth reduces it by 0.0045. These results apply to the datasets and protocol used in this study and should not be interpreted as a general claim that any facial region is uninformative.

The final 48-location set is a selection of landmark locations, not DCT coefficients. Every retained landmark continues to use the same first-12-bit coefficient prefix. The usefulness-ranked landmark list is frozen before evaluation and reused by Pipeline 3.

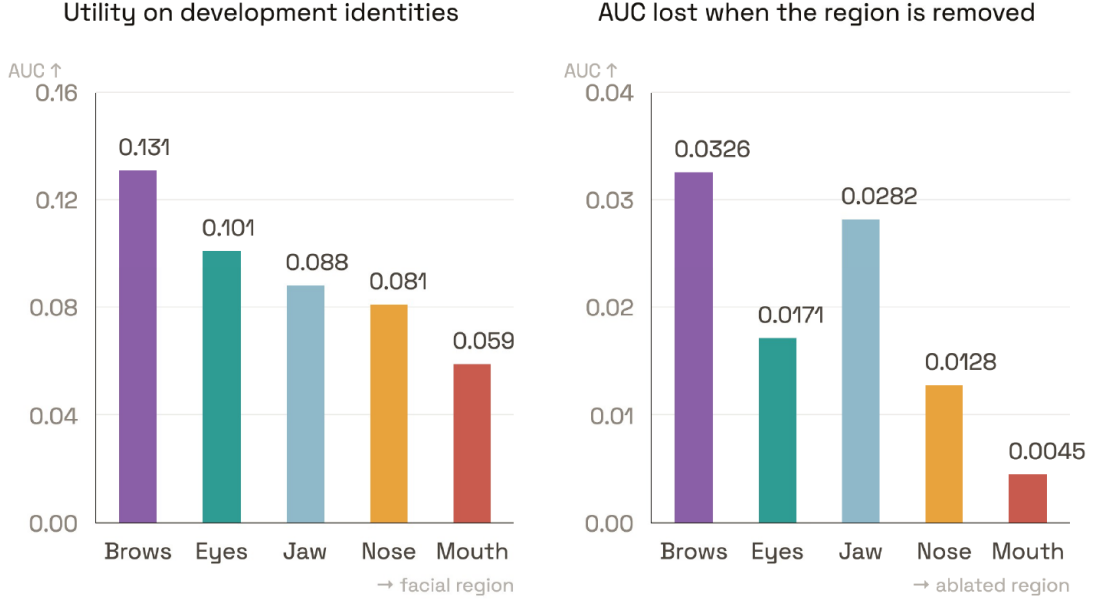


Figure 5.7. Pipeline 2 development-stage facial-region utility and ablation. Brows and eyes have the strongest average utility; removing the mouth causes the smallest AUC loss in this study.

Each selected landmark contributes one 12-bit item. DCT-LBF therefore uses 48 regional Bloom filters without any larger concatenated hierarchy levels. Let $L=48$ be the number of retained landmarks and $b=12$ the number of stored bits per landmark. The complete binary template length is

$$P_2 = L \times b = 48 \times 12 = 576 \text{ bits.} \quad (5.3)$$

and its packed byte size is

$$\left\lceil \frac{576}{8} \right\rceil = 72 \text{ bytes.} \quad (5.4)$$

One Bloom filter is allocated to each selected region for every dataset bank. The enrolled population n determines the filter length m using the target false-positive probability $f=0.01$. The accepted banks use $k=7$ hash addresses. The storage required by a region-aware bank is

$$S_{HBF} = \left\lceil \frac{L \times m}{8} \right\rceil \text{ bytes.} \quad (5.5)$$

During enrollment, every photograph contributes its 12-bit code to the Bloom filter corresponding to each retained landmark. If several photographs of one identity are enrolled, multiple valid appearances of that local region can therefore be inserted into the same filter.

During a query, each region returns either a match or a non-match. The score is

$$s_2(q) = \sum_{r=1}^{48} I_r(q), \quad (5.6)$$

where $I_r(q)$ equals one when all required addresses for region r are set. A query is accepted when $s_2(q)$ reaches the dataset-specific threshold selected using development identities.

The choice of DCT was also evaluated against other handcrafted local descriptors. For this comparison, the faces, landmark locations, local window, 8×8 sampling grid, median-based binarization, 12 stored bits per region, selected landmark set, and region-indexed HBF were kept fixed. Only the local transform was changed.

The tested frontends were DCT, Haar wavelets, raw pooled pixels, histograms of oriented gradients (HOG), Gabor responses, binarized statistical image features (BSIF), and local binary patterns (LBP). These descriptors represent frequency, gradient, filter-response, and local-texture information in different forms [11], [12], [13], [14], [15].

Frontend	Equal-dataset macro AUC
DCT	0.803
Haar wavelet	0.787
Raw pixels	0.708
HOG	0.675
Gabor	0.666
BSIF	0.650
LBP	0.633

Table 5.2. Handcrafted frontend comparison.

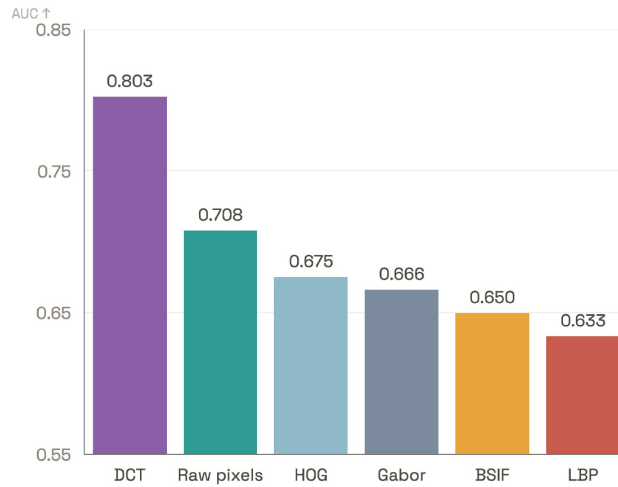


Figure 5.8. Handcrafted frontend comparison. All systems use the same faces, landmark locations, storage budget, and region-indexed HBF; only the local descriptor changes.

Among the tested handcrafted descriptors, DCT gives the highest accepted macro AUC of 0.803. Haar wavelets are next at 0.787, while the remaining methods produce lower values. This experiment is used only to select the handcrafted frontend. It is not used in place of the primary Pipeline 2 result, which follows the complete accepted protocol.

5.2.4 DCT-LBF versus Baseline HBF Results

Table 5.3 reports the DCT-LBF evaluation results against Pipeline 1: Baseline HBF. The membership thresholds are selected using development identities and frozen before evaluation. The detailed sensitivity and selectivity analysis is presented in Section 6.5 as part of the final comparison between pipelines.

Dataset	AUC	Accuracy	Correct / total	HBF-bank storage (bytes)
Faces94	0.9901	0.9140	765 / 837	29,220

Dataset	AUC	Accuracy	Correct / total	HBF-bank storage (bytes)
Faces95	0.9440	0.8473	333 / 393	13,230
Faces96	0.8863	0.7545	627 / 831	29,046
FEI	0.7981	0.7222	65 / 90	3,912
FERET	0.7192	0.7403	305 / 412	17,772

Table 5.3. Pipeline 2 evaluation results.

The equal-dataset macro AUC is

$$\frac{0.9901+0.9440+0.8863+0.7981+0.7192}{5} = 0.8675. \quad (5.7)$$

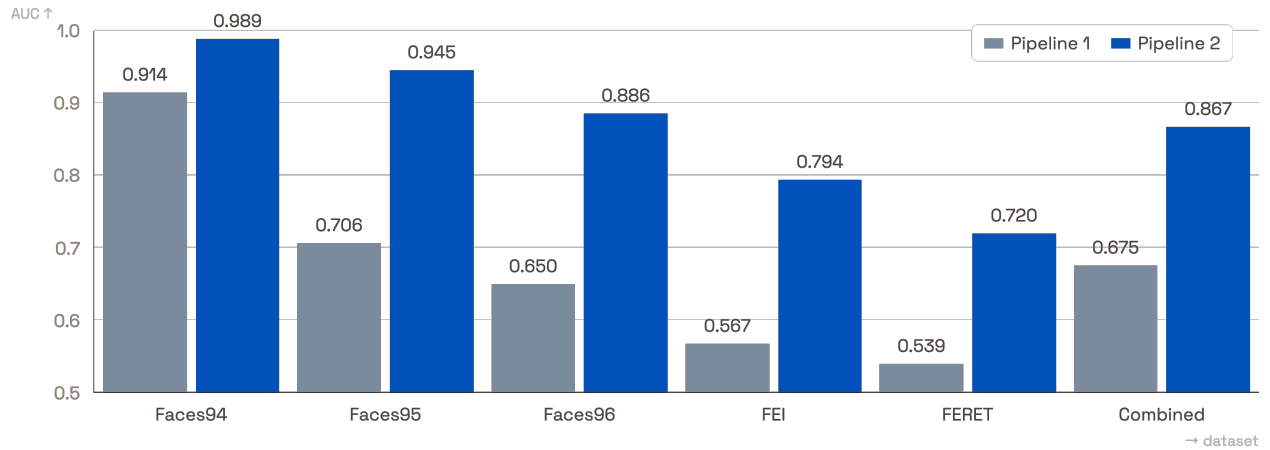


Figure 5.9. DCT-LBF AUC compared with Pipeline 1. Pipeline 2 improves every dataset and the equal-dataset macro average.

DCT-LBF improves AUC on every dataset compared with Pipeline 1. The largest improvements occur where the global pixel baseline performs most poorly. FEI increases from 0.5672 to 0.7981, while FERET increases from 0.5395 to 0.7192. Both datasets have only one enrollment photograph for each identity, so these gains are obtained without adding more enrolled appearances.

Faces94 reaches 0.9901, while Faces95 and Faces96 reach 0.9440 and 0.8863. The fixed low-frequency DCT prefix performs well across both the more controlled and the more variable image collections.

The five Pipeline 2 HBF banks require

$$29,220 + 13,230 + 29,046 + 3,912 + 17,772 = 93,180 \text{ bytes.} \quad (5.8)$$

The binary template decreases from 613 bytes to 72 bytes. The reduction is

$$100 \left(1 - \frac{72}{613} \right) = 88.3\%. \quad (5.9)$$

Across the five datasets, total HBF-bank storage decreases from 324,191 bytes to 93,180 bytes, which is a reduction of approximately 71.3%. DCT-LBF also improves selectivity relative to the baseline, even though its sensitivity to same-person bit changes is higher. The detailed e/P analysis in Section 6.3 examines how the lower impostor collision rate can still lead to improved AUC.

5.3 Deep Spatial LBF

5.3.1 Motivation

Pipeline 3, Deep Spatial LBF, is the second implementation of the proposed LBF framework. It keeps the same landmark-indexed Bloom-filter organization as DCT-LBF, but replaces the handcrafted DCT descriptor with spatial features taken from a frozen ArcFace face-recognition network.

Pipeline 3 is intended to test whether the local representation can be improved while keeping the LBF structure unchanged. DCT-LBF already showed that organizing the Bloom filters around facial landmarks can improve recognition and reduce storage. The remaining question is whether a stronger identity-oriented descriptor can improve the stability of those local codes.

Shomaji et al. propose LBP, PCA, and CNN-based inputs as possible directions for improving the HBF representation [2]. Pipeline 3 examines the CNN direction with an identity-trained frontend while retaining the landmark-indexed organization introduced in Pipeline 2.

ArcFace is used here as a controlled frontend rather than as the contribution by itself. During training, ArcFace uses an additive angular-margin loss that encourages samples from the same identity to remain closer while separating different identities in angular space [6]. The ArcFace network is not trained or fine-tuned as part of this study. The pretrained model is frozen and used only to extract features.

FaceNet is another valid face-recognition approach and learns a Euclidean embedding through triplet loss and online triplet mining [29]. A full comparison between face-recognition backbones is outside the scope of this study. The selected model is the reproducible InsightFace buffalo_1 ResNet50@WebFace600K checkpoint [17]. Its IResNet-50 implementation provides a $512 \times 7 \times 7$ spatial feature tensor that can be sampled at the facial-landmark locations. Building a comparable FaceNet spatial frontend would require selecting both an implementation and an intermediate feature layer, which would add another variable to the comparison.

The purpose of Pipeline 3 is therefore to test one controlled substitution. The 48 development-selected landmarks, 12-bit code per landmark, 576-bit (72-byte) template, 48 region-indexed Bloom filters, membership-score definition, identity-disjoint partitions, and development-only threshold selection remain the same as in Pipeline 2. Only the local descriptor used to produce each regional code is changed. This allows the effect of identity-trained spatial features to be measured without increasing the storage budget or changing the LBF backend.

5.3.2 Architecture and Methodology

Pipeline 3 begins with the same common preprocessing used by the earlier pipelines. The face is detected and 68 facial landmarks are estimated. From these landmarks, five points are used for ArcFace alignment: the center of the left eye, the center of the right eye, the nose tip, and the two mouth corners. Let the detected points be a_i and the standard ArcFace target points be t_i for $i=1, \dots, 5$.

A least-squares similarity transform estimates scale, rotation, and translation between the detected points and the standard ArcFace locations. The transform maps the face into a 112×112 ArcFace input image. The same transformation is also applied to all 68 facial landmarks so that their coordinates remain consistent with the transformed image.

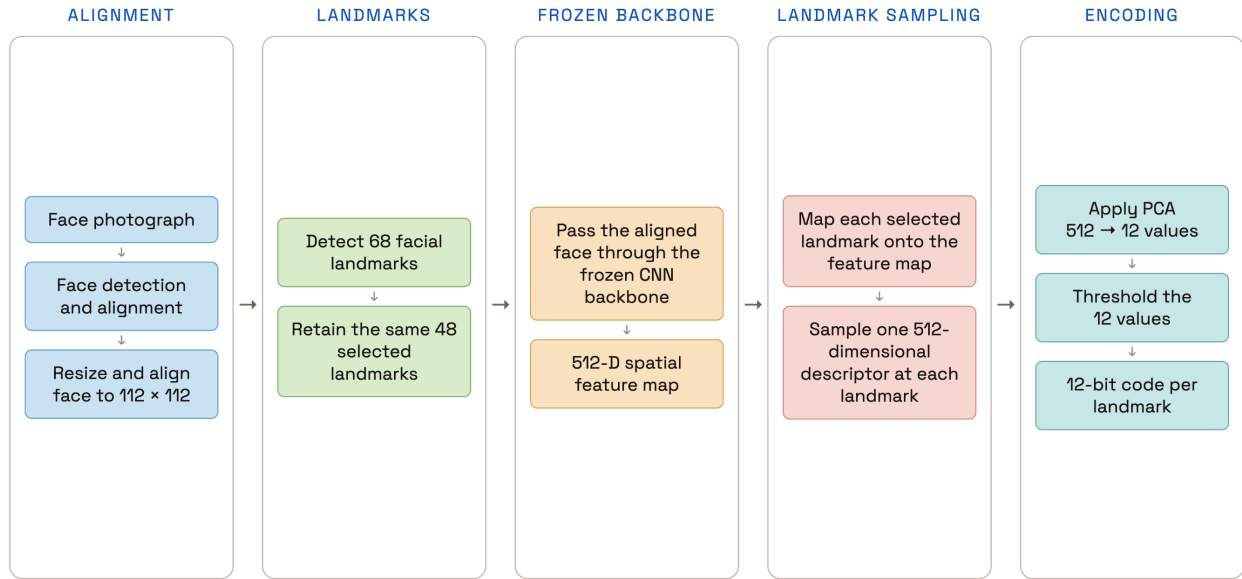


Figure 5.10. Pipeline 3 architecture. PCA whitening and seeded rotation are folded into one fitted 512×12 matrix; the 72-byte template and region-indexed HBF remain fixed.

This second alignment has a different role from the common 160×160 alignment used earlier.

The common alignment gives all pipelines a consistent starting point. The ArcFace alignment places the face into the coordinate system expected by the pretrained recognition network.

Figure 5.10 summarizes the complete Deep Spatial LBF architecture.

Figure 5.11 shows how the frozen ArcFace network produces a spatial feature map and how the selected landmarks sample local descriptors. It illustrates the processing sequence and is not an additional experiment.

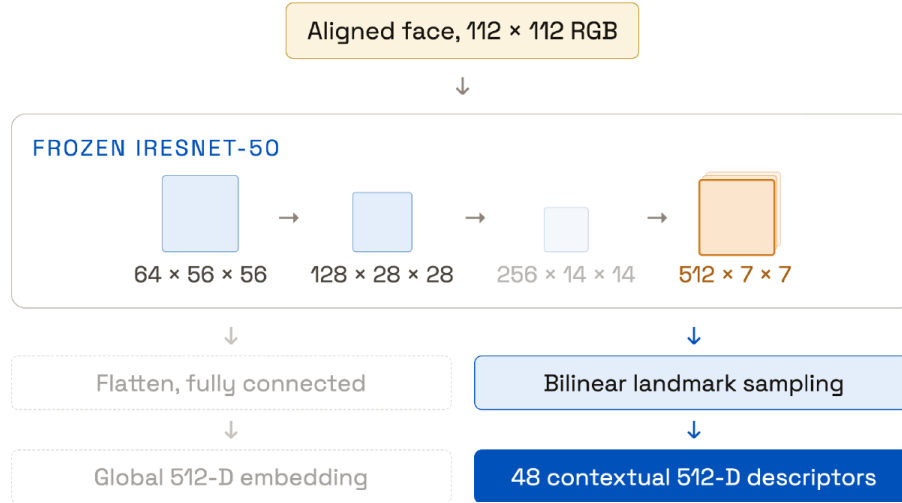


Figure 5.11. Pipeline 3 (Deep Spatial LBF) spatial feature-map extraction and landmark sampling. The frozen network retains a 7×7 spatial map, and each selected landmark samples one local 512-dimensional descriptor.

The ArcFace frontend uses the InsightFace recognition checkpoint w600k_r50.onnx, which is a ResNet-50 model trained on WebFace600K [17]. A SHA-256 checksum is recorded for the local checkpoint. The network weights remain frozen and are not trained or fine-tuned using any of the thesis datasets.

The normal ArcFace output is a 512-dimensional whole-face embedding [6]. Deep Spatial LBF does not use only this final embedding. Instead, it reads the batch-normalization output from the final residual stage so that some spatial structure is retained. For one input image, this feature tensor has shape

$$512 \times 7 \times 7. \quad (5.10)$$

The first dimension contains 512 feature channels, while the remaining 7×7 dimensions describe coarse spatial locations in the face. Only one forward pass through the network is needed for each image. The same spatial tensor is then used to obtain descriptors for all landmark locations, so the network does not need to be evaluated separately for every landmark. Because the 68 landmarks were transformed using the same similarity transform as the face, their coordinates now refer to the 112×112 ArcFace image. These image coordinates must then be mapped to the 7×7 feature map.

For a landmark coordinate x and feature-map width $W=7$, the continuous horizontal feature-map coordinate is

$$g_x = (x + 0.5) \frac{W}{112} - 0.5. \quad (5.11)$$

The vertical coordinate g_y is calculated in the same way. If a mapped location falls outside the feature map, it is clamped to the nearest border. Bilinear interpolation is then used to combine values from the surrounding feature-map cells.

Sampling at one landmark produces a 512-dimensional real-valued descriptor. The sampling location is tied to a particular facial landmark, but the feature channels have been generated by a deep network with a broad receptive field. The descriptor is therefore sampled locally while still containing contextual information from a larger part of the face.

A 512-dimensional real-valued descriptor is much larger than the 12-bit regional code allowed by the LBF design. Pipeline 3 uses PCA as a linear compression step so that the ArcFace descriptor can be placed into the same storage budget used by DCT-LBF. The projection retains the main directions of variation observed in fit identities, which is more suitable for a fixed 12-bit code than choosing twelve channels by index.

Pipeline 3 fits a PCA-based linear encoder using only fit identities [16]. The encoder compresses each 512-dimensional landmark descriptor directly to 12 values, matching the fixed 12-bit regional budget used by Pipeline 2. All fitted settings are frozen before evaluation.

At runtime, the frozen encoder produces 12 values for each selected landmark. Each value is converted to one bit using a zero threshold: positive values become 1, while zero or negative values become 0.

The same 48 development-selected landmarks are retained, producing the same 576-bit (72-byte) template used by DCT-LBF. The encoder and zero threshold are fixed before evaluation, and no evaluation identity is used to choose the projection or binary encoding.

5.3.3 One Controlled Substitution

Deep Spatial LBF changes only the descriptor used at each landmark. The 48 selected landmark locations, 12 bits per landmark, 576-bit (72-byte) template, region-indexed LBF backend, HBF-bank storage configuration, thresholds, and identity-disjoint evaluation protocol are all retained from DCT-LBF. The comparison therefore measures the effect of replacing the handcrafted local descriptor with an identity-trained one, rather than changing the size of the template or the membership bank.

The Pipeline 3 backend is intentionally the same size as Pipeline 2. At each of the 48 landmarks, the 12 sign bits form one regional block. The complete template size is

$$P_3 = 48 \times 12 = 576 \text{ bits} = 72 \text{ bytes.} \quad (5.12)$$

The block name and binary value are processed using the same keyed HMAC-SHA-256 and FNV-1a address construction used by the other pipelines [18], [19], [20]. Each of the 48 Bloom filters is sized according to the enrolled population of the dataset, with a target false-positive probability of $f=0.01$ and $k=7$ hash addresses.

For a query q , the membership score is

$$s_3(q) = \sum_{r=1}^{48} l_r(q). \quad (5.13)$$

The threshold is selected separately for each dataset using development identities and is then frozen before evaluation. The score retains the same interpretation as in DCT-LBF: it is the number of selected facial regions whose local code is found in the corresponding enrolled Bloom filter.

5.3.4 Deep Spatial LBF versus DCT-LBF Results

Table 5.4 reports the evaluation results for Pipeline 3.

Dataset	AUC	Accuracy	Correct / total	HBF-bank storage (bytes)
Faces94	0.9944	0.9486	794 / 837	29,220
Faces95	0.9963	0.9491	373 / 393	13,230
Faces96	0.9303	0.7834	651 / 831	29,046
FEI	0.9198	0.8333	75 / 90	3,912
FERET	0.8264	0.8010	330 / 412	17,772

Table 5.4. Pipeline 3 evaluation results.

The equal-dataset macro AUC is

$$\frac{0.9944+0.9963+0.9303+0.9198+0.8264}{5} = 0.9334. \quad (5.14)$$

Deep Spatial LBF produces the strongest recognition result on every evaluated dataset. Faces94 reaches an AUC of 0.9944, Faces95 reaches 0.9963, and Faces96 reaches 0.9303. On the one-image enrollment datasets, FEI reaches 0.9198 and FERET reaches 0.8264. Compared with DCT-LBF, these correspond to improvements of 0.1217 AUC on FEI and 0.1072 AUC on FERET.

These gains are obtained without changing the selected landmarks, the 12-bit regional code size, or the 72-byte template. The LBF backend and HBF-bank storage also remain unchanged.

Pipeline 3 also gives the highest accuracy on all five datasets under their frozen dataset-specific thresholds. Chapter 6 presents the final comparison among all pipelines together with the sensitivity and selectivity analysis.

Chapter 6: Results and Discussion

6.1 Evaluation Comparison Framework

The comparison follows the same identity-disjoint primary protocol, frozen operating thresholds, landmark windows, Bloom-filter parameters, and backend decision rule for all three pipelines.

Pipeline 1 is the global-pixel HBF baseline, Pipeline 2 replaces the global descriptor with landmark-indexed DCT regions, and Pipeline 3 changes only the regional frontend to ArcFace spatial features. This ordering separates the effect of landmark indexing from the effect of the learned ArcFace representation.

6.2 Overall Three-Pipeline Results

Table 6.1 gives the headline recognition and storage results. Macro AUC and macro accuracy are equal-dataset arithmetic means; the accuracy values use each dataset's frozen development threshold. Storage is reported both per protected template and for the five retained HBF banks.

Pipeline	Macro AUC	Macro accuracy	Template (bytes)	Five HBF banks (bytes)
Pipeline 1	0.6754	0.6751	613	324,191
Pipeline 2	0.8675	0.7957	72	93,180
Pipeline 3	0.9334	0.8631	72	93,180

Table 6.1. Overall recognition and storage summary.

Across the complete progression, macro AUC rises from 0.6754 to 0.9334, an increase of 25.81 percentage points (38.21% relative). Macro accuracy rises from 0.6751 to 0.8631, an increase of

18.80 points (27.85% relative). At the same time, the protected template falls from 613 to 72 bytes (88.25% smaller), and the five-bank footprint falls from 324,191 to 93,180 bytes (71.26% smaller).

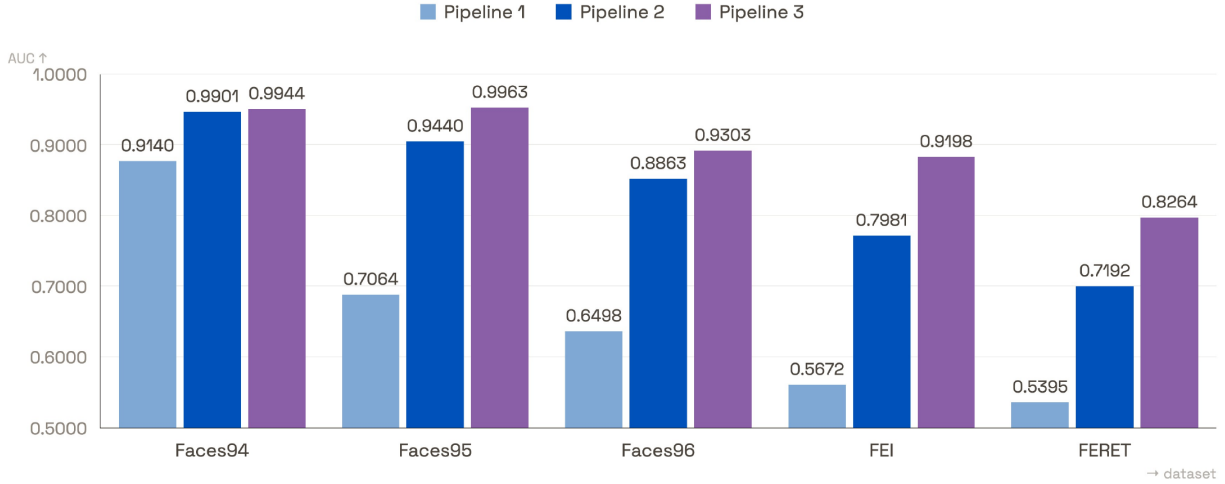


Figure 6.1. Per-dataset AUC for all three pipelines under the frozen primary protocol.

The five-dataset view shows the same monotonic ordering on every dataset. The largest practical gains occur on FEI and FERET, where uncontrolled pose, illumination, and appearance variation expose the limitations of the global and handcrafted representations.

The results make the design trajectory explicit: Pipeline 2 produces the decisive compression, while Pipeline 3 preserves those storage gains and supplies the strongest recognition. Both proposed pipelines use the same 576-bit landmark-indexed backend.

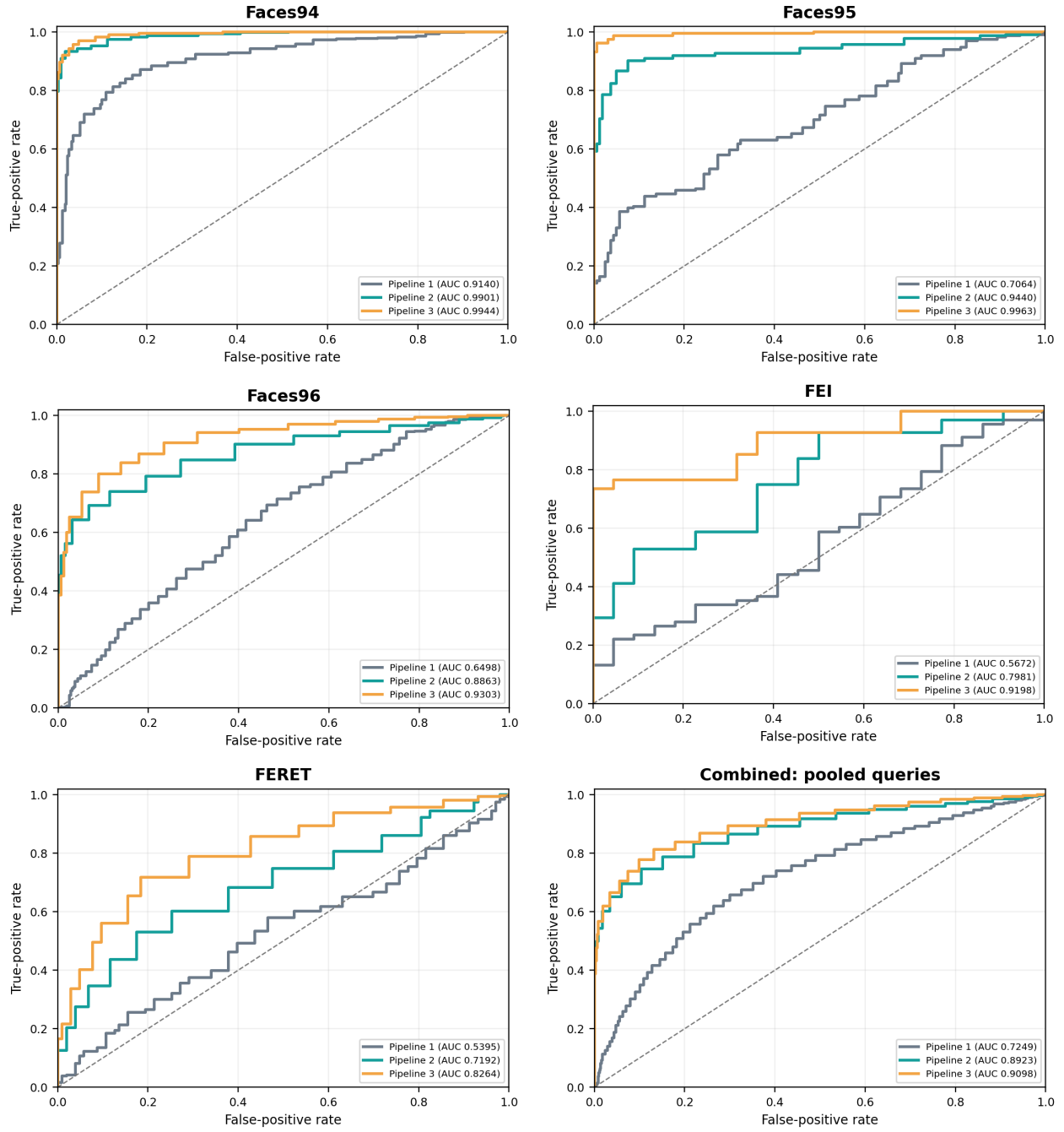


Figure 6.2. ROC-AUC curves for Pipelines 1–3 on the five datasets and the pooled combined evaluation.

The ROC curves confirm that the macro improvement is not created by a single operating threshold. Pipeline 3 generally maintains the best true-positive/false-positive trade-off across the evaluated range, followed by Pipeline 2 and then Pipeline 1.

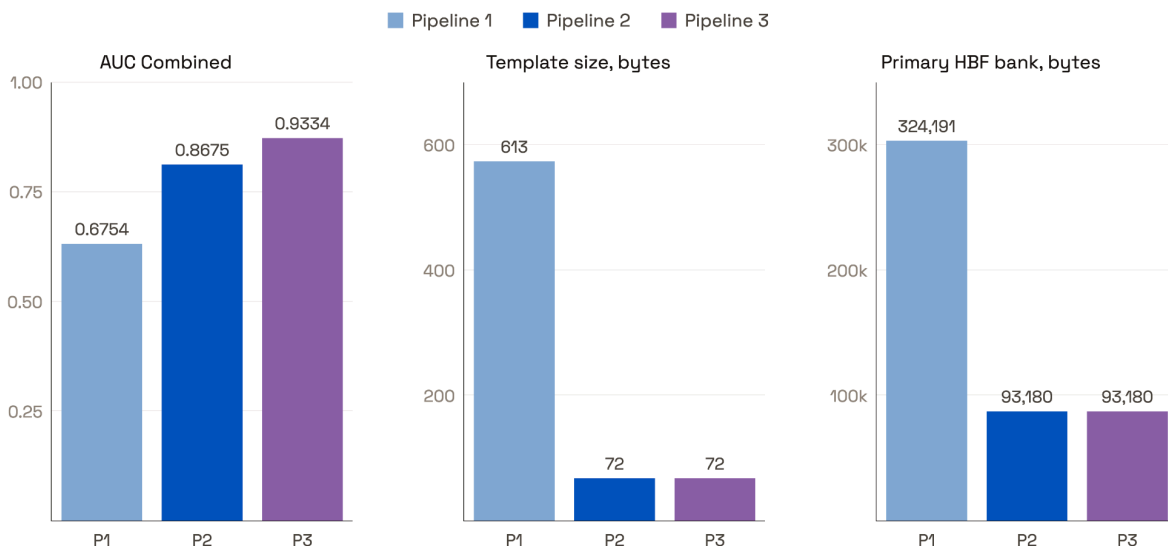


Figure 6.3. Overall recognition and storage comparison across the three pipelines. Pipeline 2 and Pipeline 3 use the same 72-byte template and 93,180-byte five-bank footprint, while Pipeline 3 gives the highest combined AUC.

6.3 Pipeline 1 to Pipeline 2: Improvements and Remaining Limitations

Moving from the global-pixel baseline to DCT-LBF raises macro AUC by 19.22 percentage points (28.45% relative) and macro accuracy by 12.06 points (17.86% relative). The protected template becomes 541 bytes smaller (88.25%), and five-bank storage falls by 231,011 bytes (71.26%). These simultaneous gains show that landmark indexing is not merely a compression step: it also places evidence into more stable, locally corresponding facial regions.

Pipeline 2 nevertheless retains a fixed handcrafted DCT prefix in every region. Its evidence ratio e/P rises from 0.1165 to 0.1551 (+33.13%), while genuine block survival changes from 0.2736 to 0.2677 (−2.16%). Although exact impostor collision falls strongly from 0.3545 to 0.1997 (−43.67%), FEI and FERET remain the weakest datasets. These results identify the remaining

problem as regional representation quality rather than Bloom-filter capacity, motivating the controlled ArcFace substitution.

6.4 Pipeline 2 to Pipeline 3: Representation Upgrade

Replacing DCT with ArcFace spatial features raises macro AUC from 0.8675 to 0.9334, a gain of 6.59 percentage points (7.60% relative), and raises macro accuracy from 0.7957 to 0.8631, a gain of 6.74 points (8.47% relative). Figure 6.3 summarizes the recognition and storage progression across all three pipelines.

The improvement is achieved with no additional protected-template or HBF-bank storage: both pipelines use 72-byte templates and 93,180-byte five-bank footprints. The gain is therefore concentrated where intended, which is in the selectivity of the regional evidence.

6.5 Sensitivity and Selectivity

The transition diagnostics explain why the recognition curves improve. The evidence ratio e/P measures the retained evidence relative to the configured query budget, genuine block survival measures the fraction of enrolled regional blocks recovered by genuine probes, and exact impostor collision measures how often impostor blocks reproduce enrolled blocks exactly.

Lower collision is desirable; survival must be interpreted together with selectivity and the final ROC operating point.

6.5.1 Pipeline 1 to Pipeline 2

Measurement	Pipeline 1	Pipeline 2	Pipeline 3	Preferred direction
Genuine e/P	0.1165	0.1551	0.1175	Lower
Genuine block survival	0.2736	0.2677	0.2992	Higher
Impostor block collision	0.3545	0.1997	0.1727	Lower

Table 6.2. Backend transition diagnostics for Pipelines 1–3.

Pipeline 2 reduces exact impostor collision by 43.67%, from 0.3545 to 0.1997, while genuine block survival changes by only -2.16% . The higher e/P ratio reflects a different distribution of regional evidence, but the large fall in collision is the more important result: impostor probes reproduce fewer enrolled regional patterns while genuine retrieval is broadly retained.

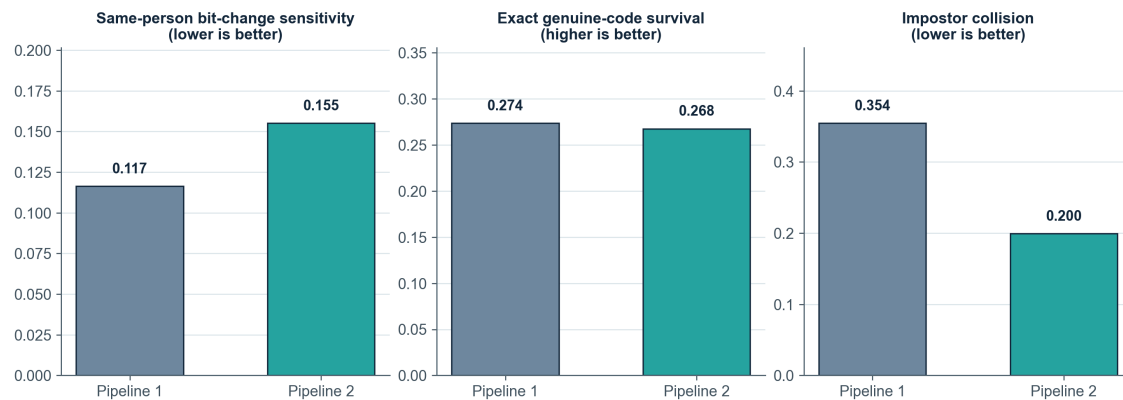


Figure 6.4. Sensitivity and selectivity diagnostics for Pipeline 1 versus Pipeline 2.

6.5.2 Pipeline 2 to Pipeline 3

Pipeline 3 lowers e/P from 0.1551 to 0.1175 (-24.24%), raises genuine block survival from 0.2677 to 0.2992 ($+11.77\%$), and lowers exact impostor collision from 0.1997 to 0.1727

(−13.52%). This is the desired combined movement: more genuine regional evidence survives while fewer impostor blocks collide.

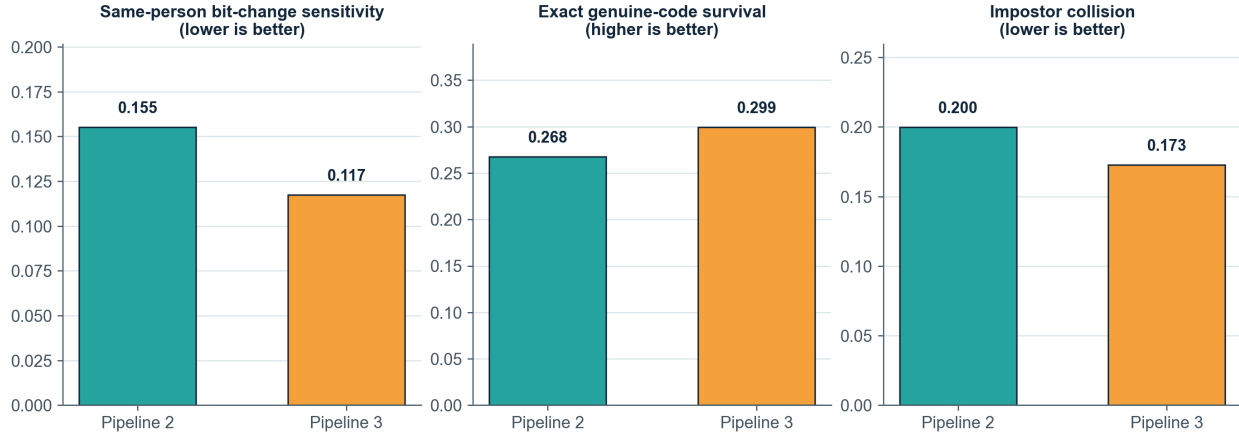


Figure 6.5. Sensitivity and selectivity diagnostics for Pipeline 2 versus Pipeline 3.

6.6 Discussion of Trade-Offs and Interpretation

The results support a staged interpretation. Pipeline 1 reveals the weakness of global pixel evidence. Pipeline 2 sharply reduces storage and impostor collision, but its fixed DCT representation leaves accuracy on the table. Pipeline 3 retains the same compact regional backend and uses ArcFace spatial features to improve genuine survival and lower collision further. No single metric is sufficient; the conclusion follows from recognition, storage, and transition diagnostics moving together.

The protected templates should not be described as irreversible solely because they are hashed. The implementation uses a keyed HMAC-SHA256 step before non-cryptographic Bloom indexing, but formal unlinkability, revocability, entropy, and inversion-resistance tests remain future work under ISO/IEC 24745-style criteria [30]. The reported security-related evidence is therefore empirical collision behavior under the stated protocol, not a proof against inversion or cross-database linkage.

Chapter 7: Conclusion and Future Work

7.1 Conclusion

This thesis studied how to improve a Hierarchical Bloom Filter for face membership verification. The published HBF provides a useful way to store and query noisy biometric blocks [2], but its original face representation uses a global mean-thresholded pixel template. The work began by implementing and auditing that baseline, then used diagnostic evidence to design two compact region-aware pipelines.

Pipeline 1 reaches a macro AUC of 0.6754. DCT-LBF replaces global pixels with the first 12 low-spatial-frequency DCT bits at each retained facial landmark and reaches 0.8675. Pipeline 3 keeps the same Landmark-Indexed Bloom Filter (LBF) and uses ArcFace spatial features, reaching 0.9334. Both proposed pipelines reduce the stored binary template from 613 bytes to 72 bytes and reduce the five-bank HBF storage from 324,191 bytes to 93,180 bytes.

The baseline diagnosis and final LBF comparison led to three connected conclusions.

The baseline diagnosis shows that the middle and top hierarchy levels provide little useful ranking information because long blocks rarely survive variation between photographs. Useful membership evidence is not distributed evenly across facial regions. Changing Bloom-filter size has only a small effect once the blocks are formed. Many pixel-template positions and low-level blocks are constant across enrolled identities, so genuine and impostor queries can produce similar matches. The baseline limitation is therefore mainly a representation and organization problem. The HBF backend can store and query blocks correctly, but it cannot create identity information that is missing from the pixel code.

The region-aware results show that local DCT and ArcFace spatial features can improve membership verification while reducing storage. Pipeline 2 and Pipeline 3 improve AUC on every evaluated dataset while using 88.3% smaller binary templates and approximately 71.3% smaller HBF banks than Pipeline 1. Pipeline 3 has the strongest AUC and per-dataset accuracy throughout the evaluation.

The study contribution is the Landmark-Indexed Bloom Filter framework and its two evaluated pipelines.

The first contribution is a reproducible evaluation and diagnosis of the published Baseline HBF implementation [2]. The second is evidence that hierarchy levels and facial regions do not contribute equally. The third is a region-indexed HBF that assigns one compact filter to each selected facial landmark. The fourth is the DCT-LBF, which uses a handcrafted local frequency representation. The fifth is the Deep Spatial LBF, which uses identity-trained spatial features while preserving the same 72-byte template. The final contribution is a unified comparison using AUC, per-dataset accuracy, storage, and transition diagnostics.

7.2 Limitations

The evaluation is limited to Faces94, Faces95, Faces96, the frontal FEI subset, and the active FERET subset. Results may change with other demographics, acquisition devices, severe pose, occlusion, lower image quality, or much larger enrolled populations.

The region-aware pipelines depend on successful face detection and landmark estimation. Their selected landmark subset is fitted to the present data and protocol. The ArcFace checkpoint is frozen, but the linear encoder is fitted from the available fit identities in two source groups.

The privacy claim is limited. The study measures recognition, template size, and HBF-bank size for all retained pipelines. It does not perform formal reconstruction, linkage, key-compromise, hill-climbing, or other adversarial evaluations.

7.3 Future Work

7.3.1 Deployment and Large-Scale Evaluation

The next evaluation should move beyond the current five datasets to larger enrolled populations, more demographic and acquisition diversity, and operationally realistic probe streams. In addition to AUC and accuracy, the study should track bank occupancy, failure-to-enroll rates, and performance under changing pose, illumination, camera quality, and enrollment size. This would test whether the ranking observed here persists when both identity count and operational variability grow.

7.3.2 Occlusion-Aware Region Gating

The region-aware design creates a natural route for handling masks and local occlusion. A reliability score could be computed for each landmark window from landmark confidence, local variance, gradient energy, or feature magnitude. Regions that are covered, badly localized, or nearly flat could then be disabled or downweighed, and the final membership score normalized by the number of reliable active regions. Thresholds and gating rules must be selected on development identities and evaluated on controlled synthetic masks as well as real occlusions so that missing evidence is not mistaken for negative identity evidence.

7.3.3 Improved Region Selection

The current landmark windows are fixed. A stronger selector could rank candidate regions by cross-session stability, genuine/impostor separation, redundancy with neighboring regions, and resilience to pose or landmark noise. Selection should be learned only from identity-disjoint development data and then frozen before testing. This would allow the pipeline to retain a compact evidence budget while discarding regions that are unstable or duplicate information already carried elsewhere.

7.3.4 Identity-Weighted Regional Evidence

The present decision rule treats retained regions uniformly, even though identity information is not distributed uniformly across the face. Future work could assign weights to regions according to their development-set discriminative density, stability, and reliability, giving more influence to identity-dense areas such as the periocular region when they are visible. Weight learning must be regularized, calibrated on unseen identities, and combined with the occlusion gate so that a normally informative but currently unavailable region does not dominate the decision.

7.3.5 Region-Specific Deep Encoders

A further extension is to train encoders for specific facial parts rather than sampling every region from one generic feature map. Lightweight periocular, nose, mouth, and contour encoders—or a shared backbone with region-specific heads—could learn features matched to the statistics of each area. Distillation and quantization would be important to preserve the storage and deployment advantages demonstrated by the landmark-indexed backend.

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